

Digital Image Processing

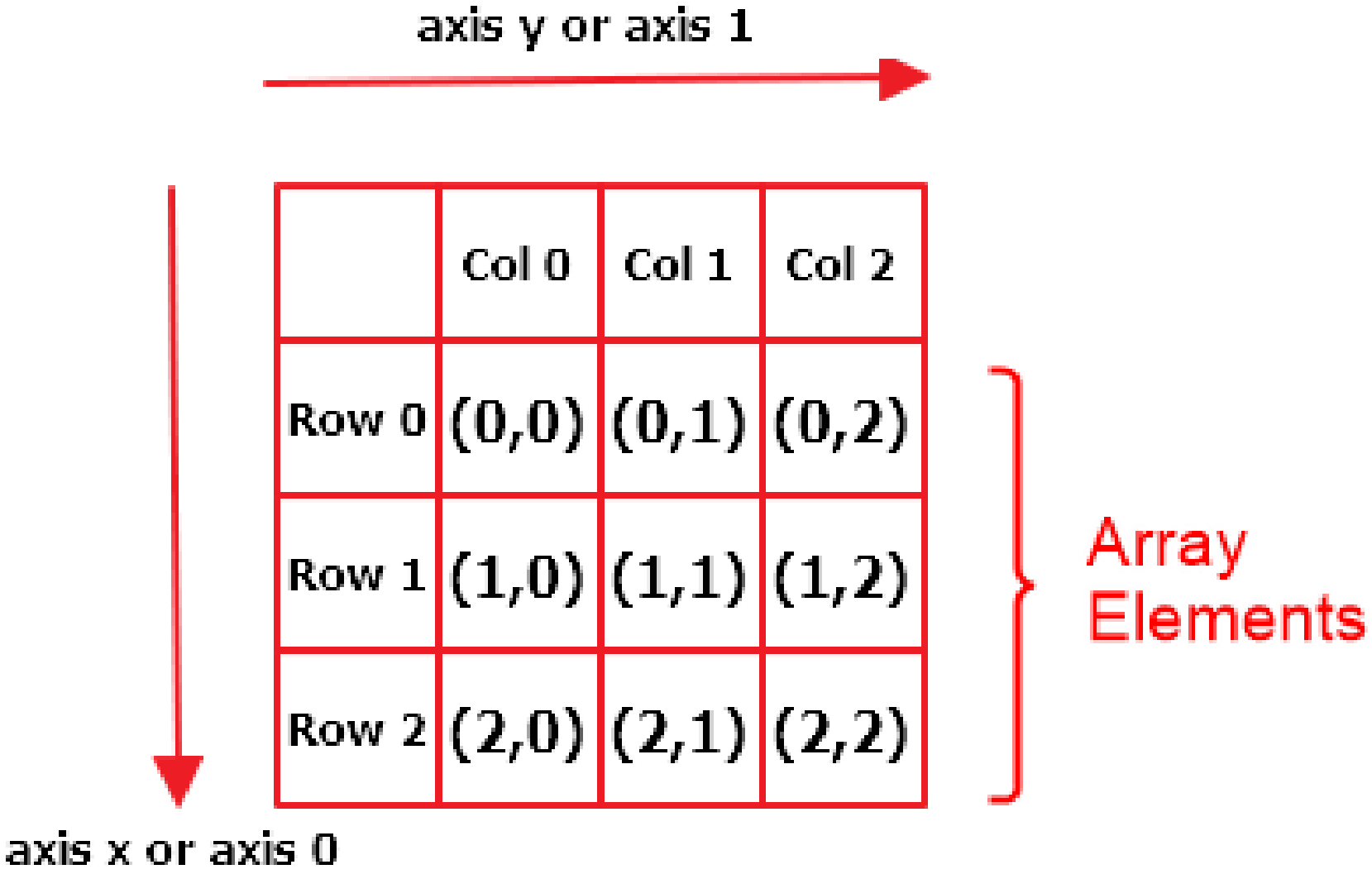
Sayed Mostafa Modaqeq
Gharjistan University
Computer Vision Workshop



WHAT IS AN IMAGE?

An image is a two-dimensional (2D) representation of a three-dimensional (3D) scene or reality.



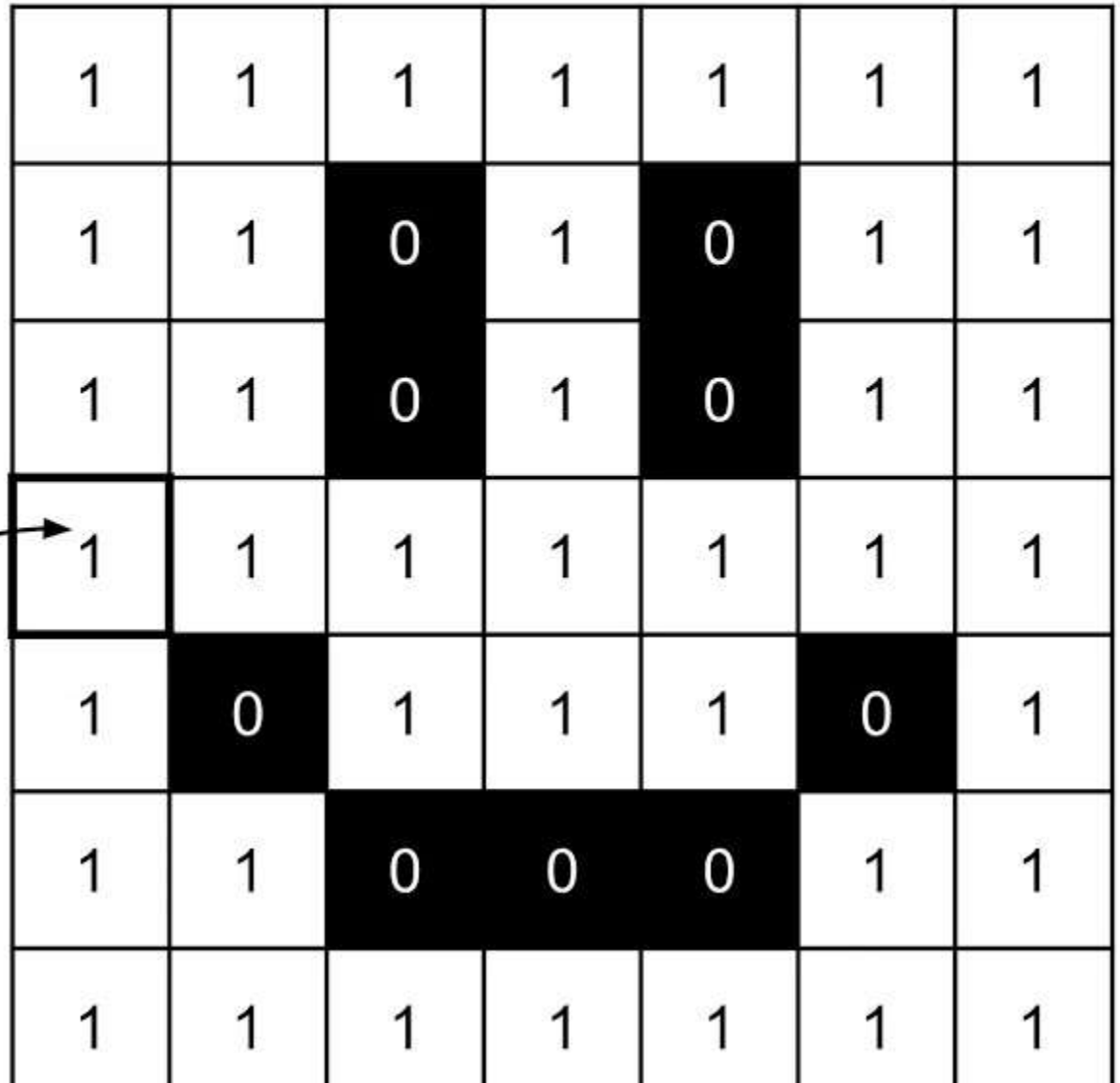


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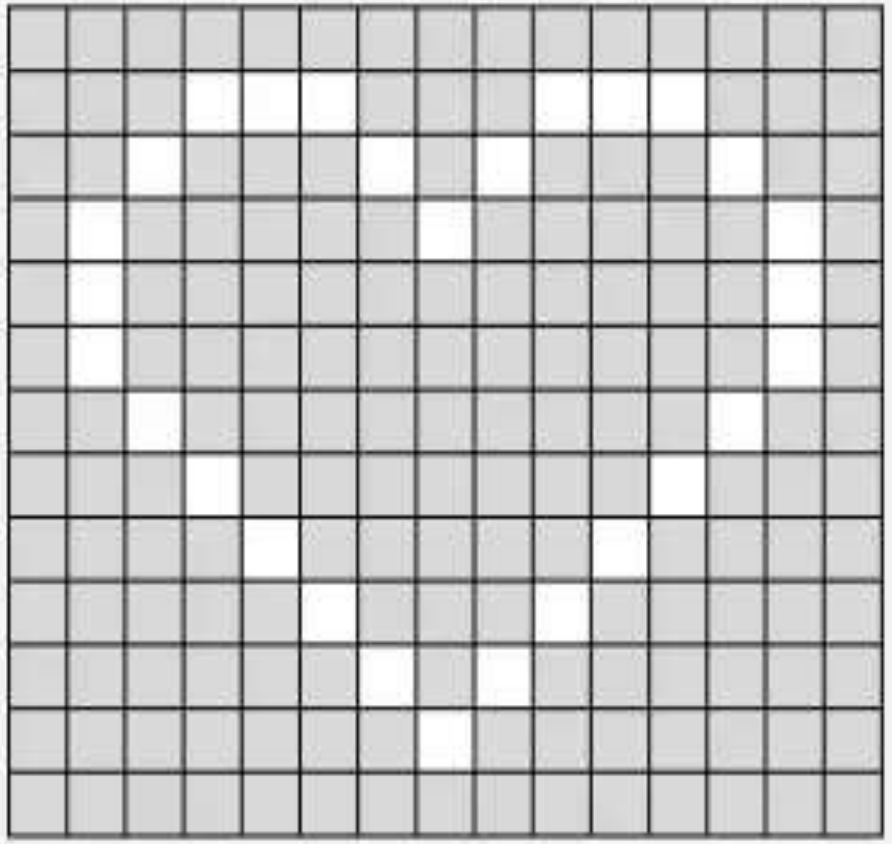
Pixel (picture element)

With only black and white we can use either 1 or 0 to represent each of these cells.

Let's be more technical though, what do we call each of these squares in an image?



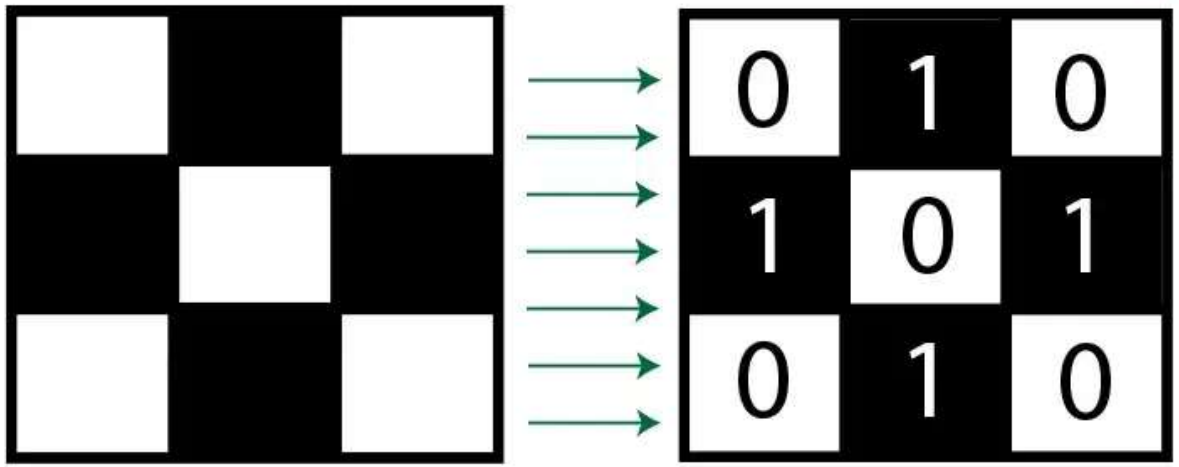
Use of matrices in image processing



$$
\begin{bmatrix}
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&1&1&1&0&0&0&1&1&1&0&0&0&0\\
0&0&1&0&0&0&1&0&1&0&0&0&1&0&0&0\\
0&1&0&0&0&0&0&1&0&0&0&0&0&1&0&0\\
0&1&0&0&0&0&0&0&0&0&0&0&0&0&1&0\\
0&1&0&0&0&0&0&0&0&0&0&0&0&0&1&0\\
0&0&1&0&0&0&0&0&0&0&0&0&0&1&0&0\\
0&0&0&1&0&0&0&0&0&0&0&0&1&0&0&0\\
0&0&0&0&1&0&0&0&0&0&0&1&0&0&0&0\\
0&0&0&0&0&1&0&0&0&1&0&0&0&0&0&0\\
0&0&0&0&0&0&1&0&1&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&1&0&0&0&0&0&0&0&0
\end{bmatrix}
$$

Basic types of images





Binary Image



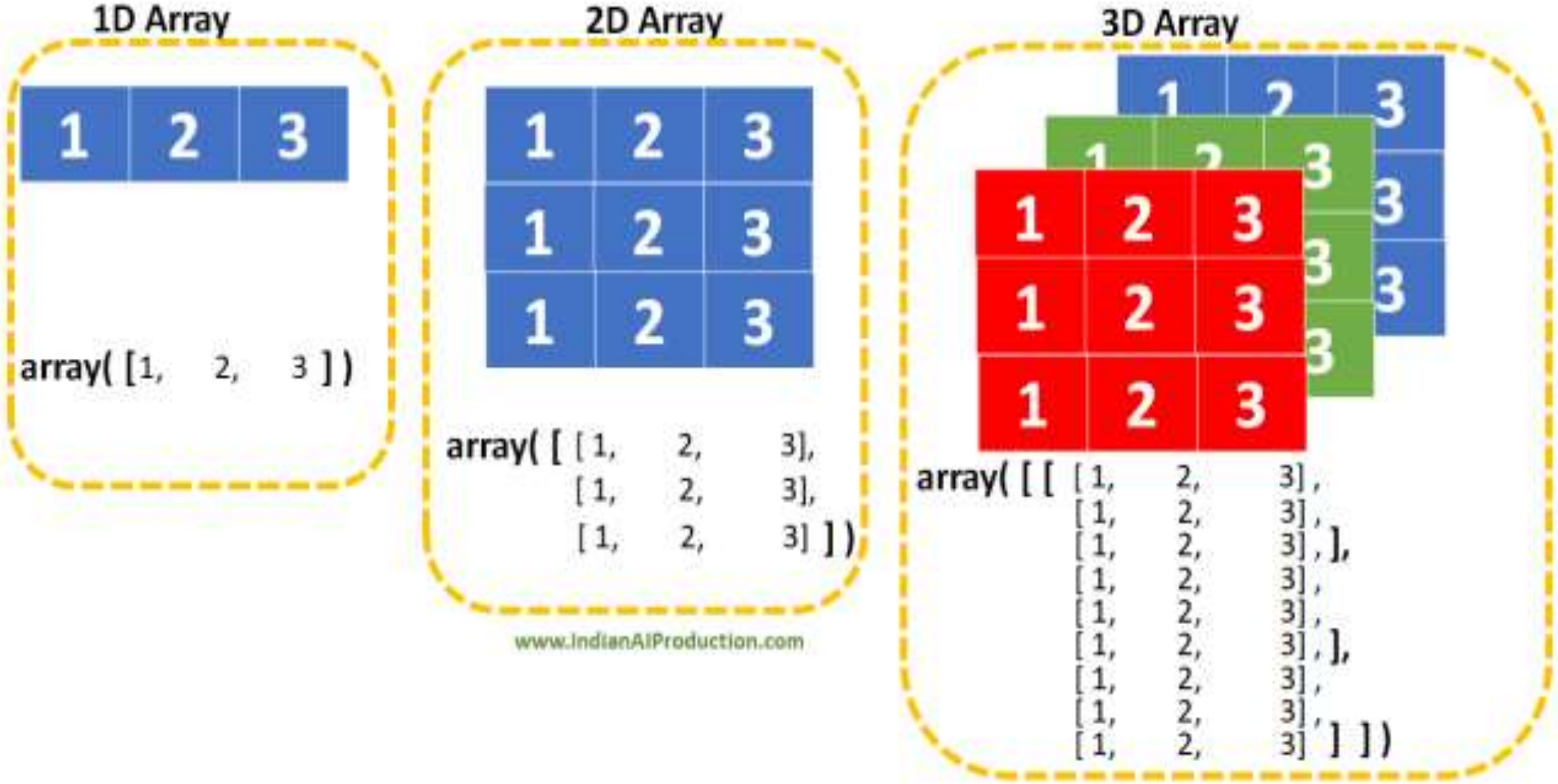
Grayscale Image



Gray image

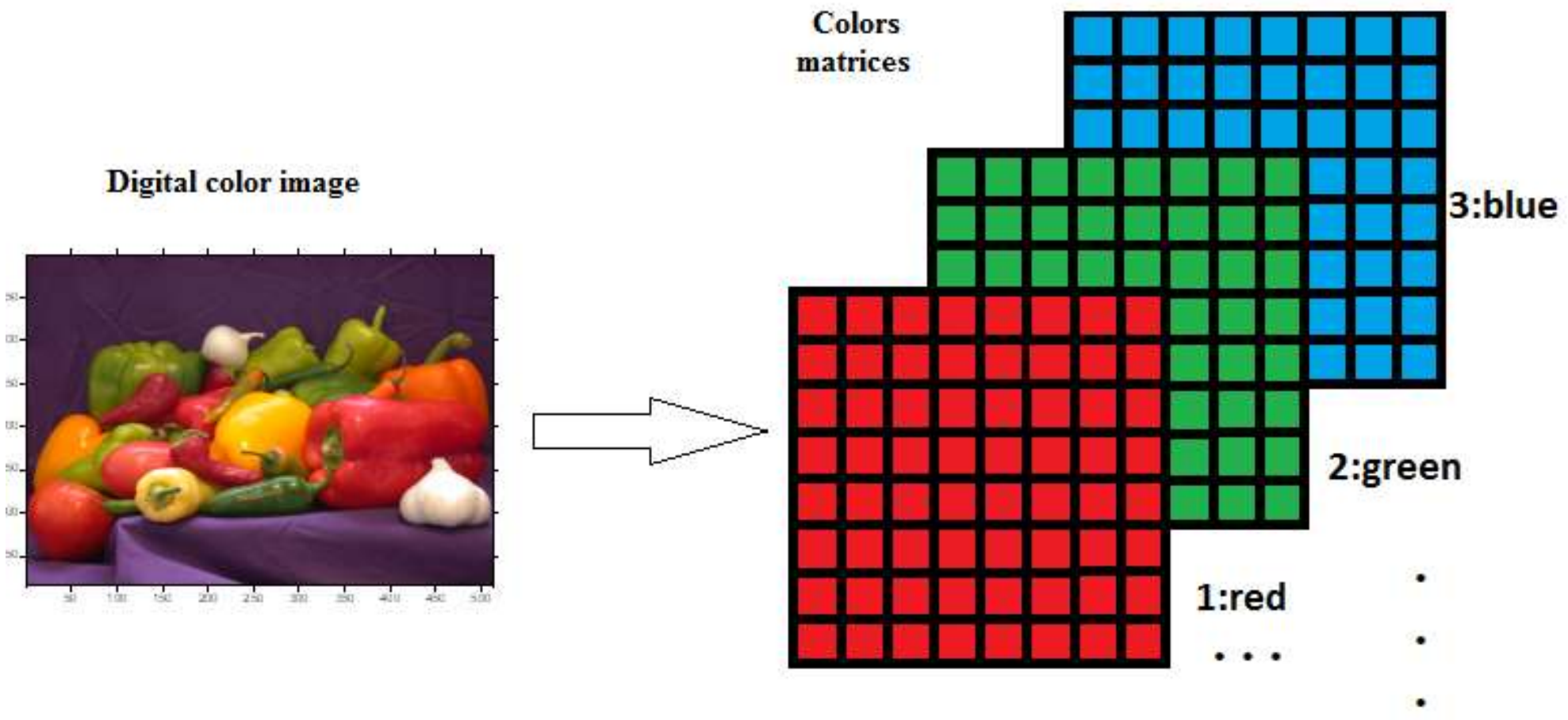
Part of the image matrix

52	80	95	103	116	131	141	145	139	138	145
42	67	75	91	100	117	142	148	145	146	148
65	114	103	129	145	145	141	132	143	154	149
81	119	107	141	150	141	141	139	165	155	146
96	117	118	155	147	138	157	151	157	152	142
92	82	140	158	139	142	152	154	162	141	133
116	77	123	136	127	143	156	150	149	139	131
120	78	116	126	122	144	159	148	139	134	132
123	90	126	135	126	143	161	152	138	130	134
144	100	131	137	129	146	168	158	137	130	134
146	102	120	126	130	153	171	156	130	130	131



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Color Image (RGB)



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a) Binary Image

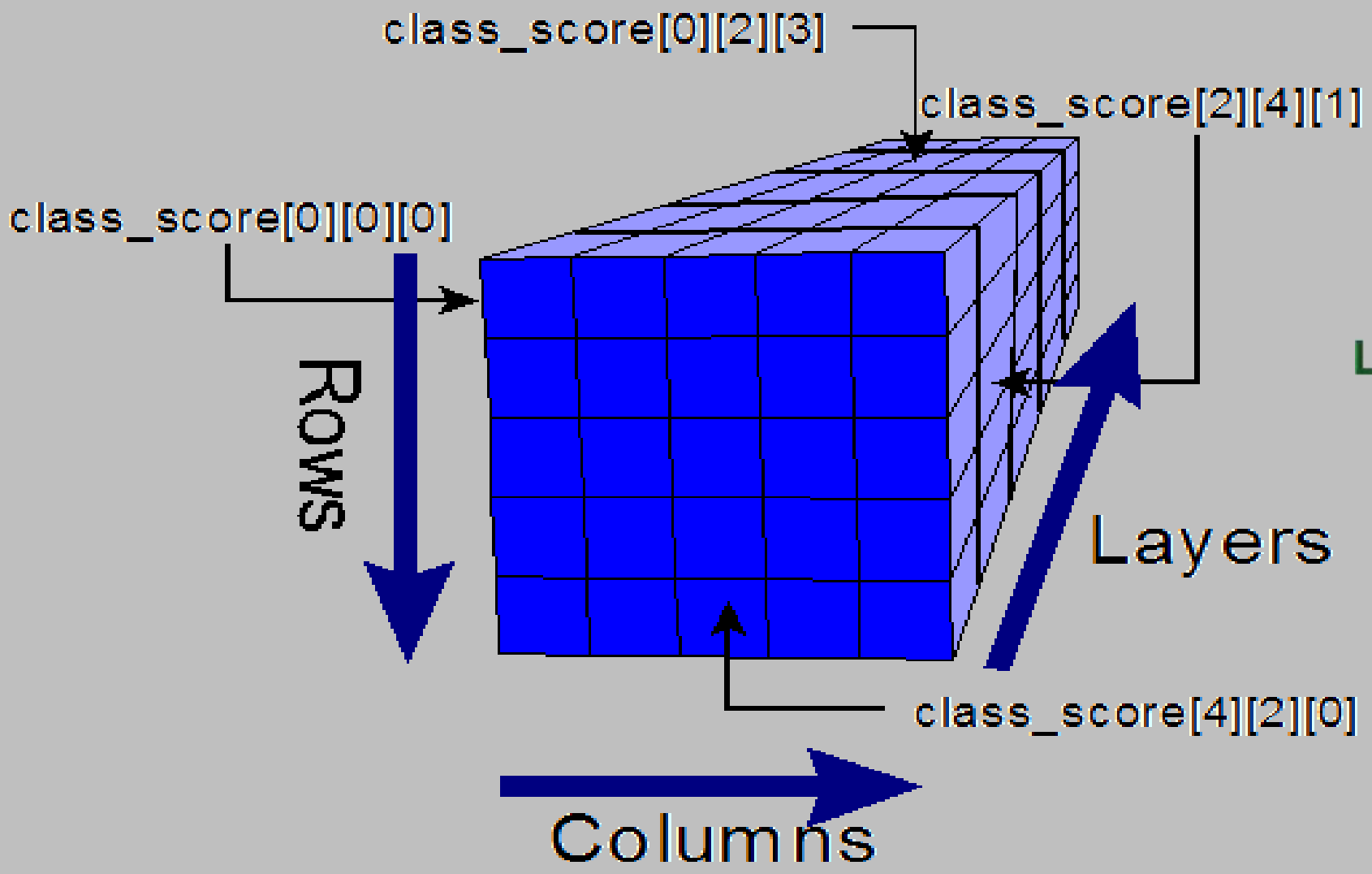


b) Gray-Scale Image



c) Color Image

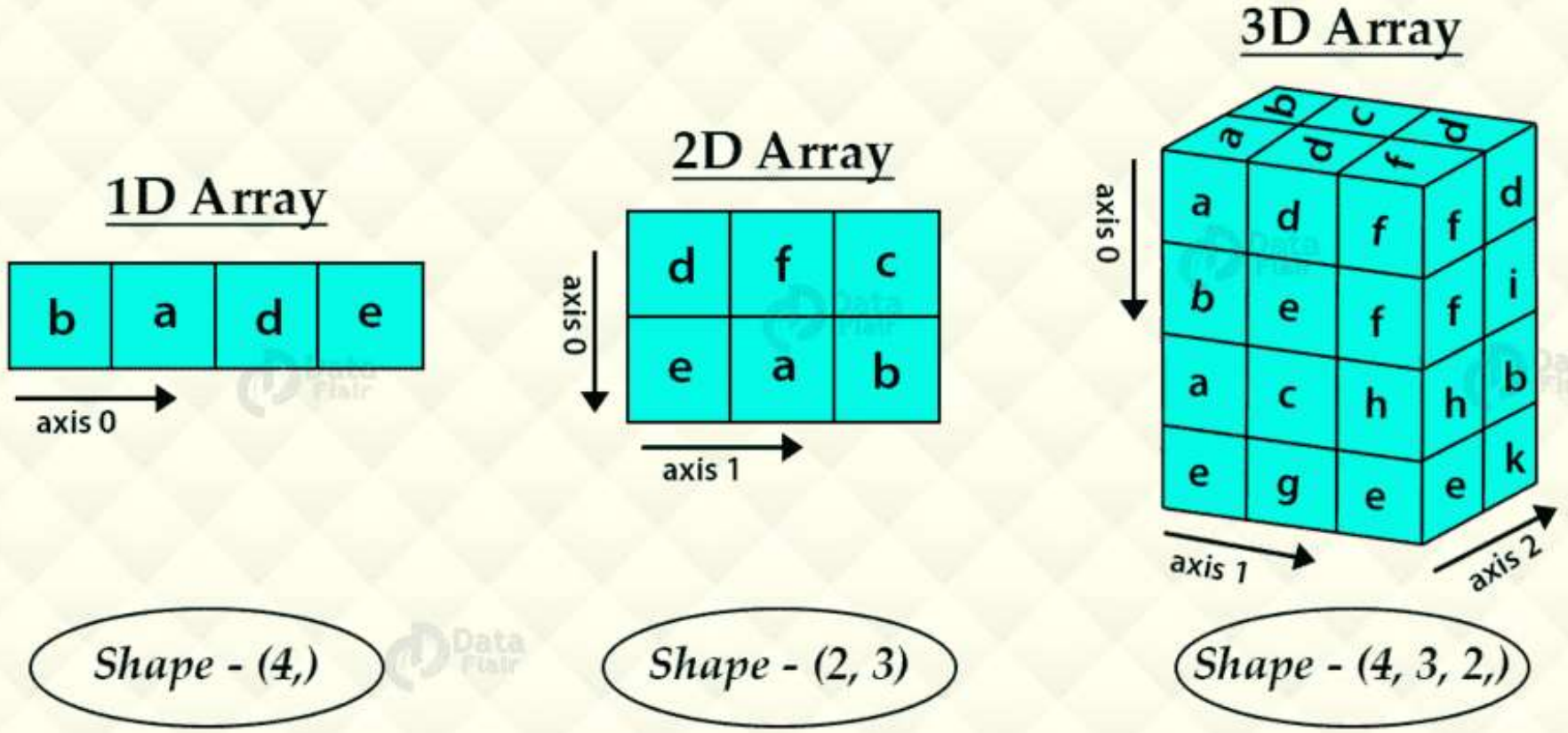
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Rows = Height
Columns = Width
Layers = Depth or Channel

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NumPy Broadcasting



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RGB to Grayscale

```
For Each Pixel in Image
        Set Red = Pixel.Red
        Set Green = Pixel.Green
        Set Blue = Pixel.Blue
                Set Gray = (Red * 0.299 +
                Green * 0.587 + Blue * 0.114)
        Set Pixel.Red = Gray
        Set Pixel.Green = Gray
        Set Pixel.Blue = Gray
End For
```




Image Enhancement





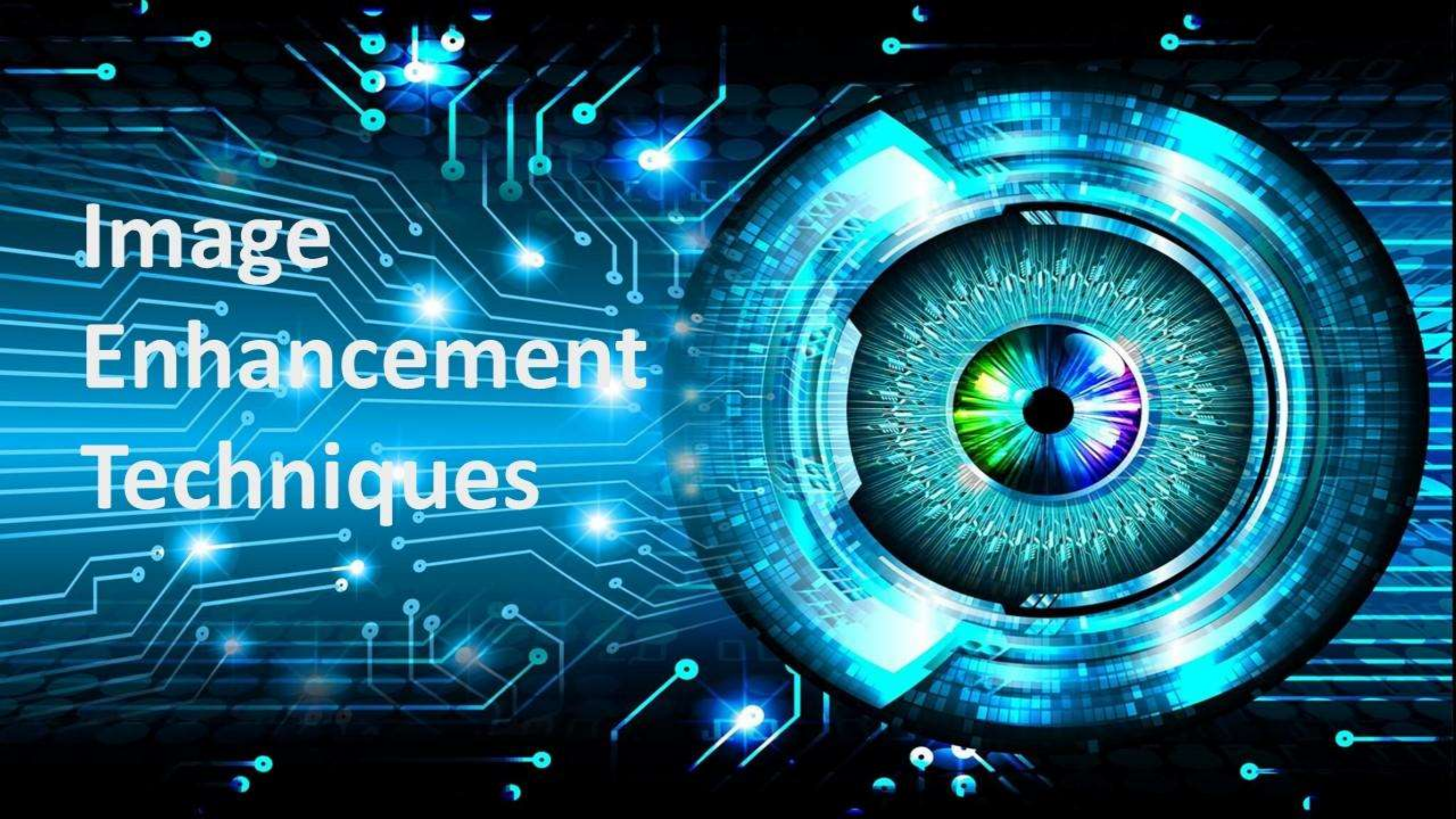
Image Enhancement

Image enhancement is basically the improving the perception of information in images for human viewers and providing better input for the further image processing techniques.

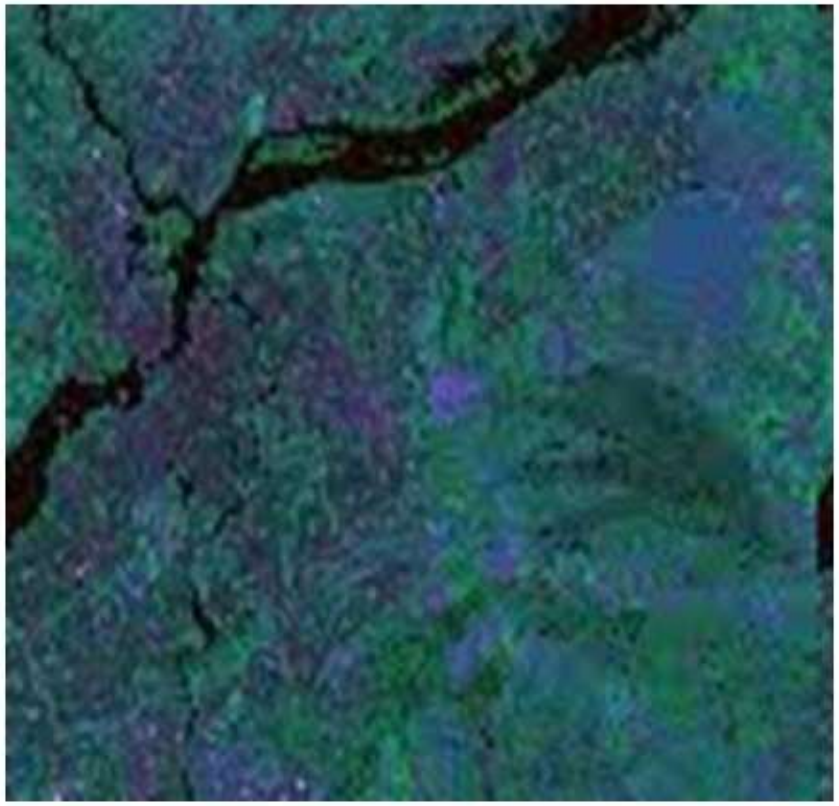
Problems

- _Poor contrast
- _Noisy
- _Wrong size and shape

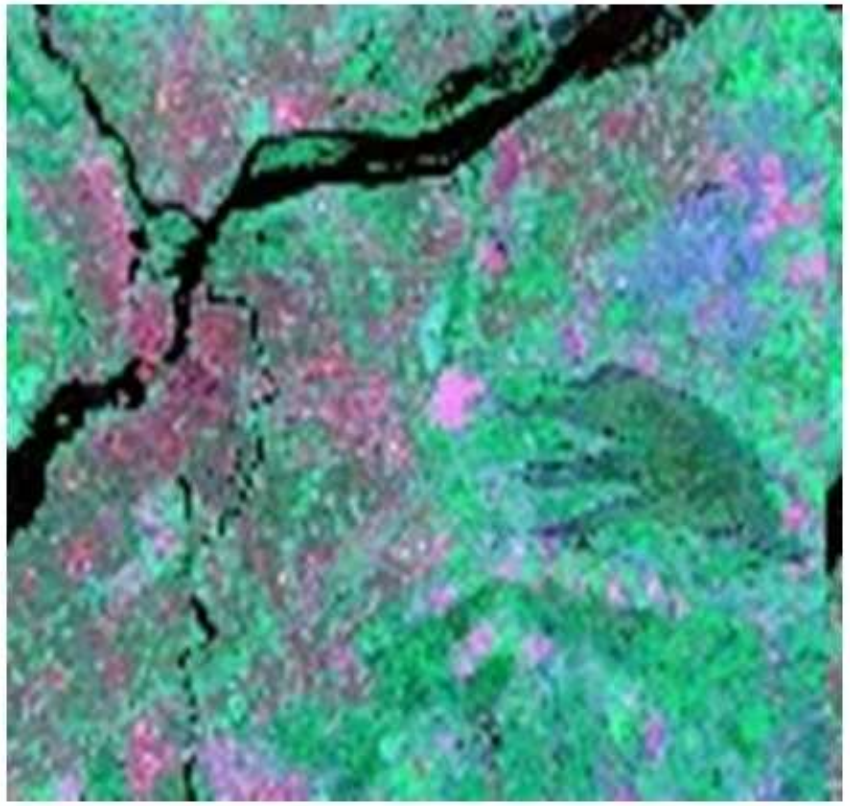
Image Enhancement Techniques



Contrast Enhancement



Before Contrast Enhancement



After Contrast Enhancement

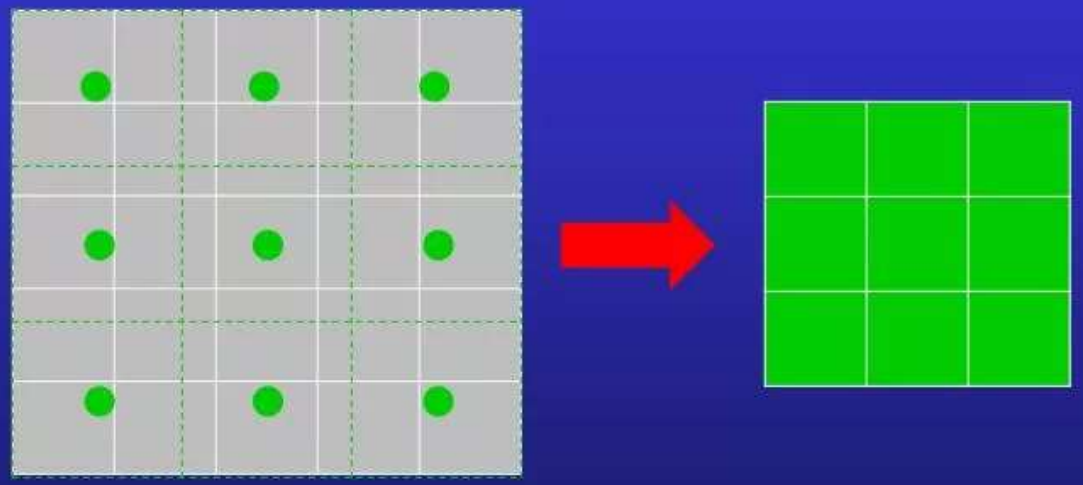
Denoising



Resampling

Image Resampling

Example: Downscaling from 5×5 to 3×3 pixels

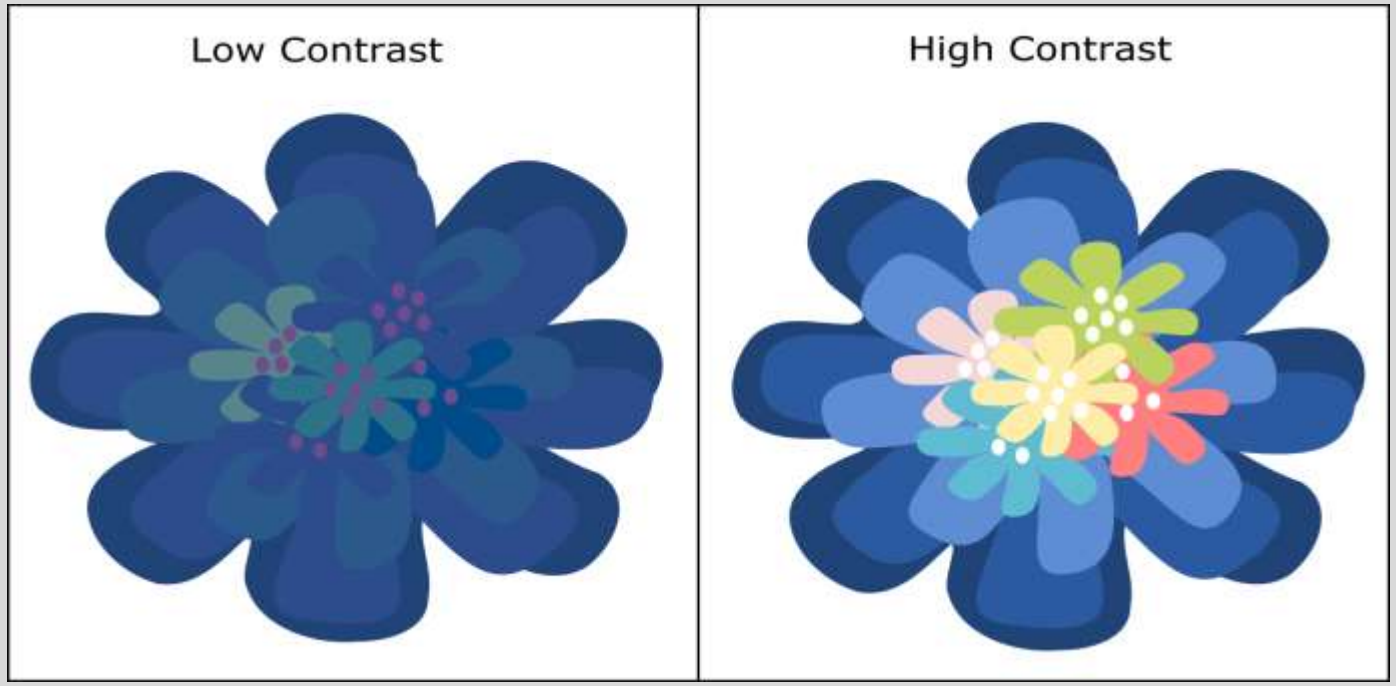


- Centers of output pixels mapped onto input image

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Contrast

Difference between the brightest and darkest parts of an image.



- Low contrast: Image looks dull and flat.
- High contrast: Image looks vivid and clear.


HIGH
CONTRAST
VS
LOW
CONTRAST



LOW CONTRAST



BALANCED

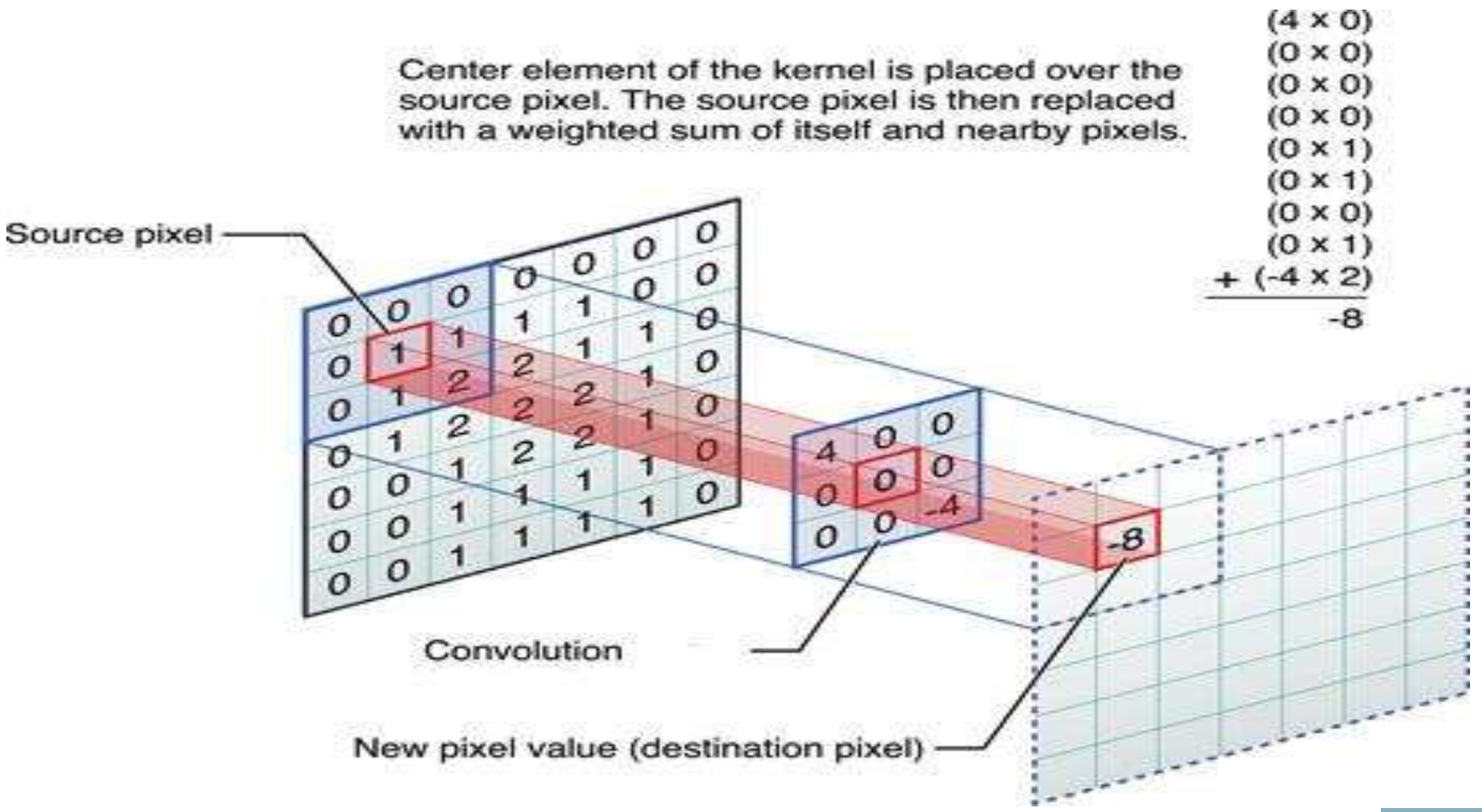


HIGH CONTRAST

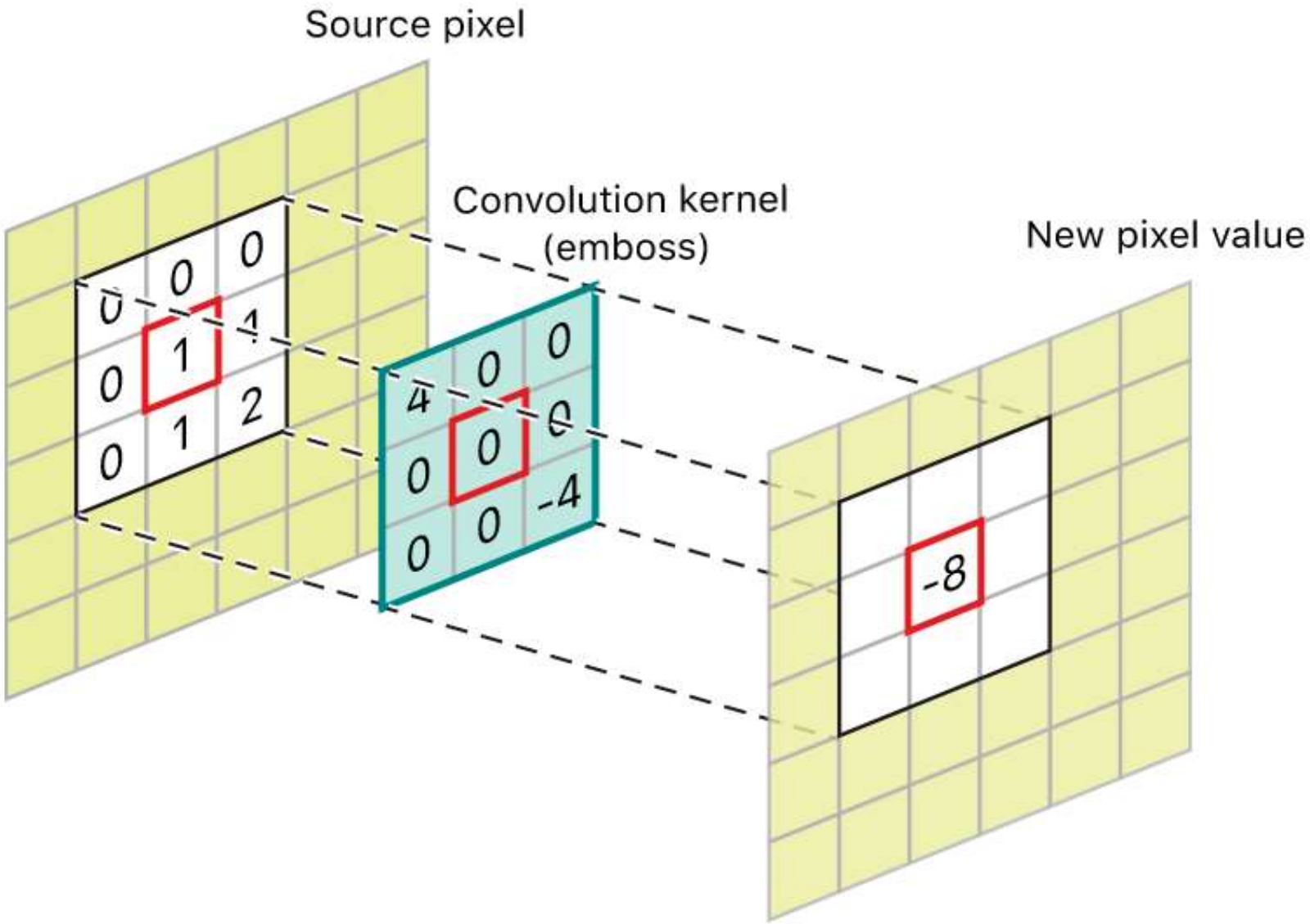
$$contrast = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$$

$$RMS = \sqrt{\frac{1}{MN} \sum_{i=1}^{N} \sum_{j=1}^{M} (Iij - avg(I))^2}$$

Convolution

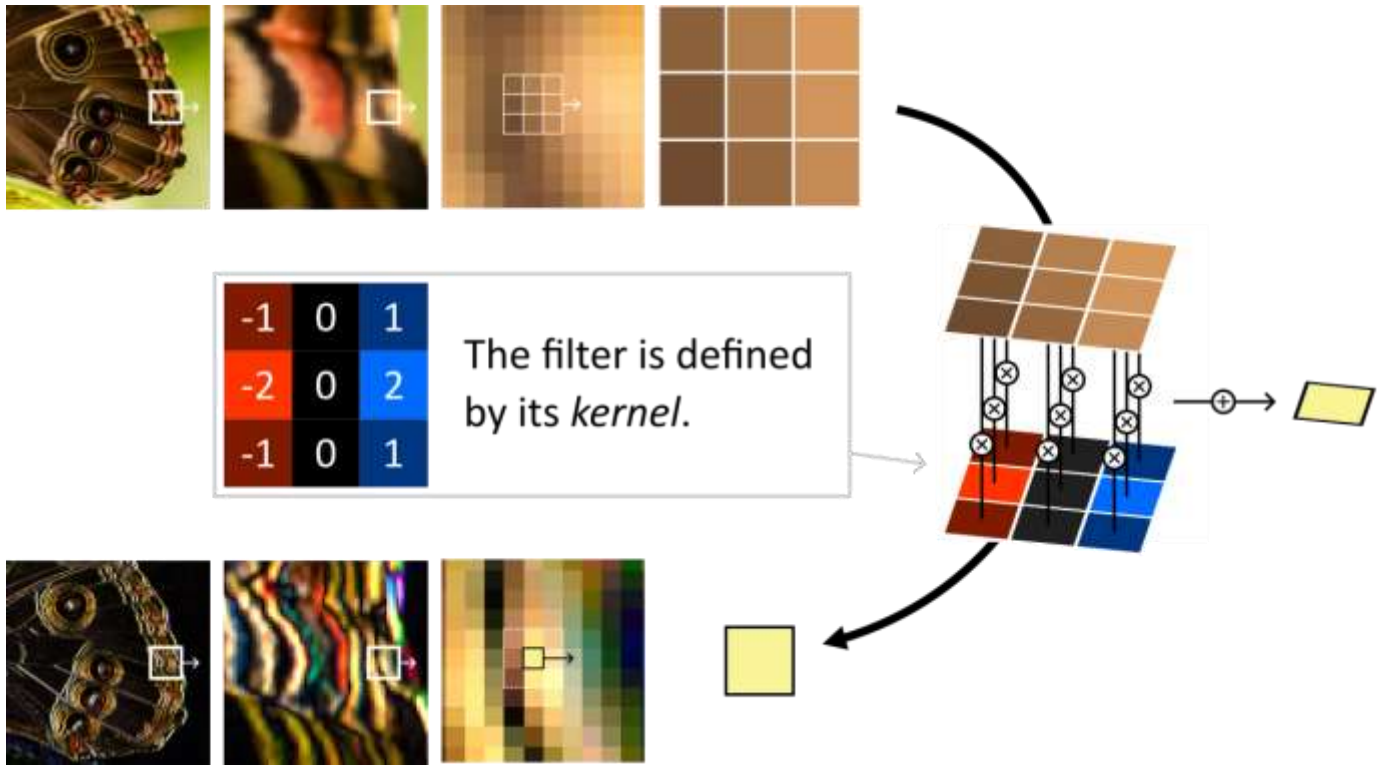


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Filter



0	0	0	0	0
0	0	1	2	0
0	3	4	5	0
0	6	7	8	0
0	0	0	0	0

*

0	1
2	3

=

0	3	8	4
9	19	25	10
21	37	43	16
6	7	8	0

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0	0	0	0	0	0	0
0	60	113	56	139	85	0
0	73	121	54	84	128	0
0	131	99	70	129	127	0
0	80	57	115	69	134	0
0	104	126	123	95	130	0
0	0	0	0	0	0	0

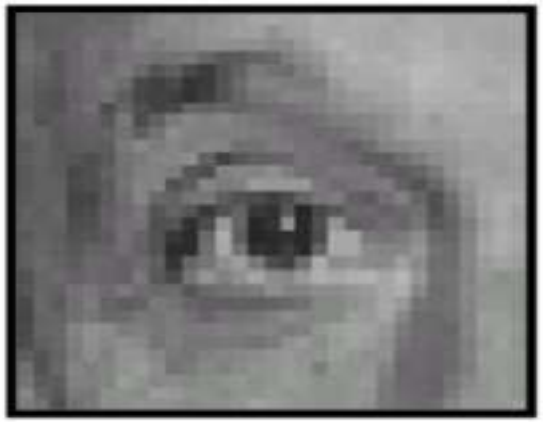
Kernel

0	-1	0
-1	5	-1
0	-1	0

114				

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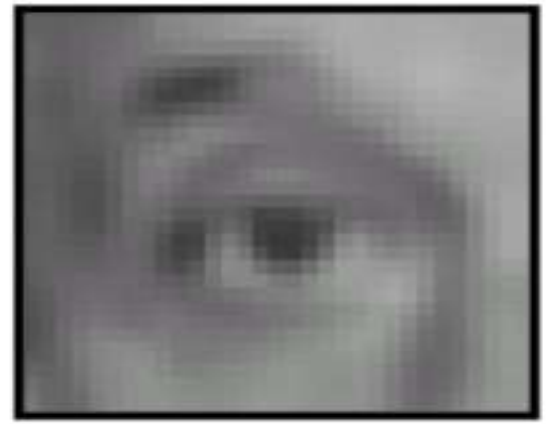


Original

$*$ $\frac{1}{9}$

1	1	1
1	1	1
1	1	1

$=$



Blur (with a mean filter)

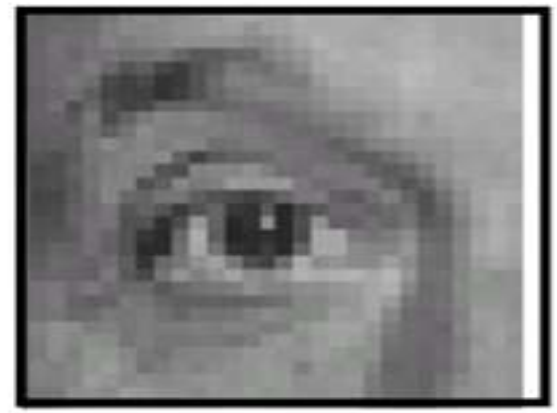


Original

$*$

0	0	0
1	0	0
0	0	0

$=$



Shifted left
By 1 pixel

Noise in Images

Definition:
Unwanted random variations in pixel values
that degrade image quality.

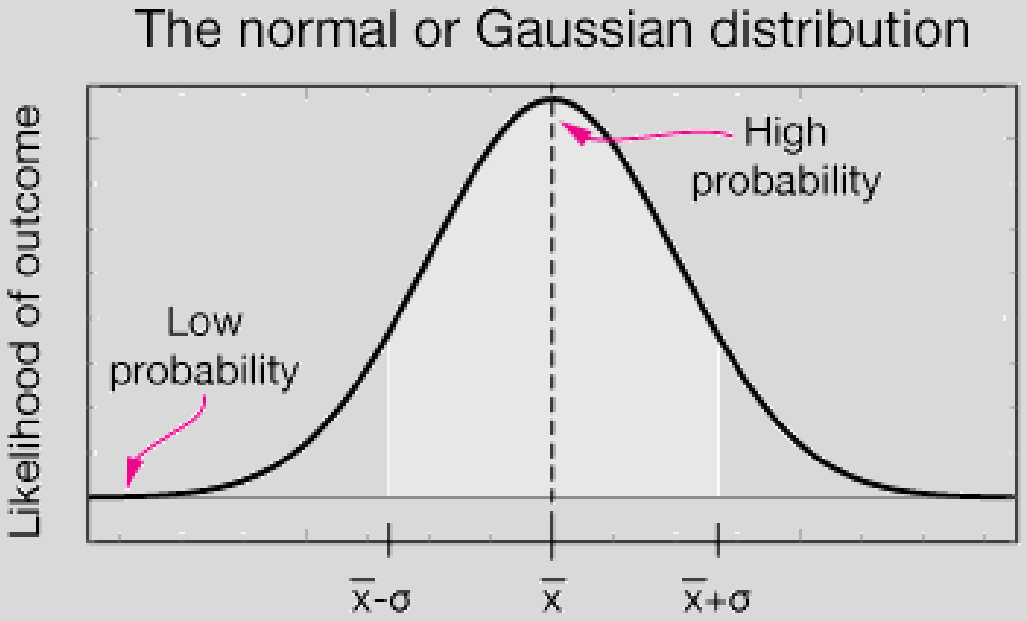
Sources:
_Sensor errors
_transmission
_compression



TYPES OF NOISE IN IMAGE PREPROCESSING

1.Gaussian




The normal or Gaussian distribution
Likelihood of outcome
High probability
Low probability
x̄-σ
x̄
x̄+σ


(a)
(b)

2.Salt and Pepper







3.Speckle

SPECKLE NOISE

- Speckle noise can be modeled by **random values multiplied** by pixel values of an image.



Fig 1: Original Image

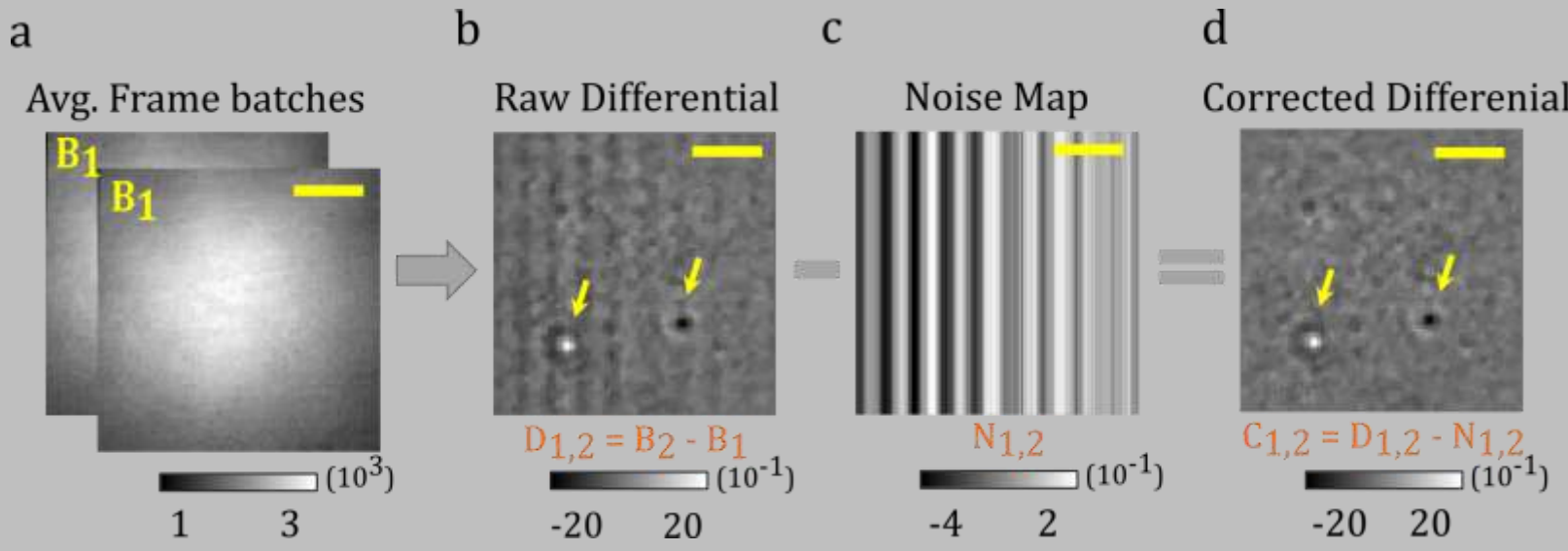


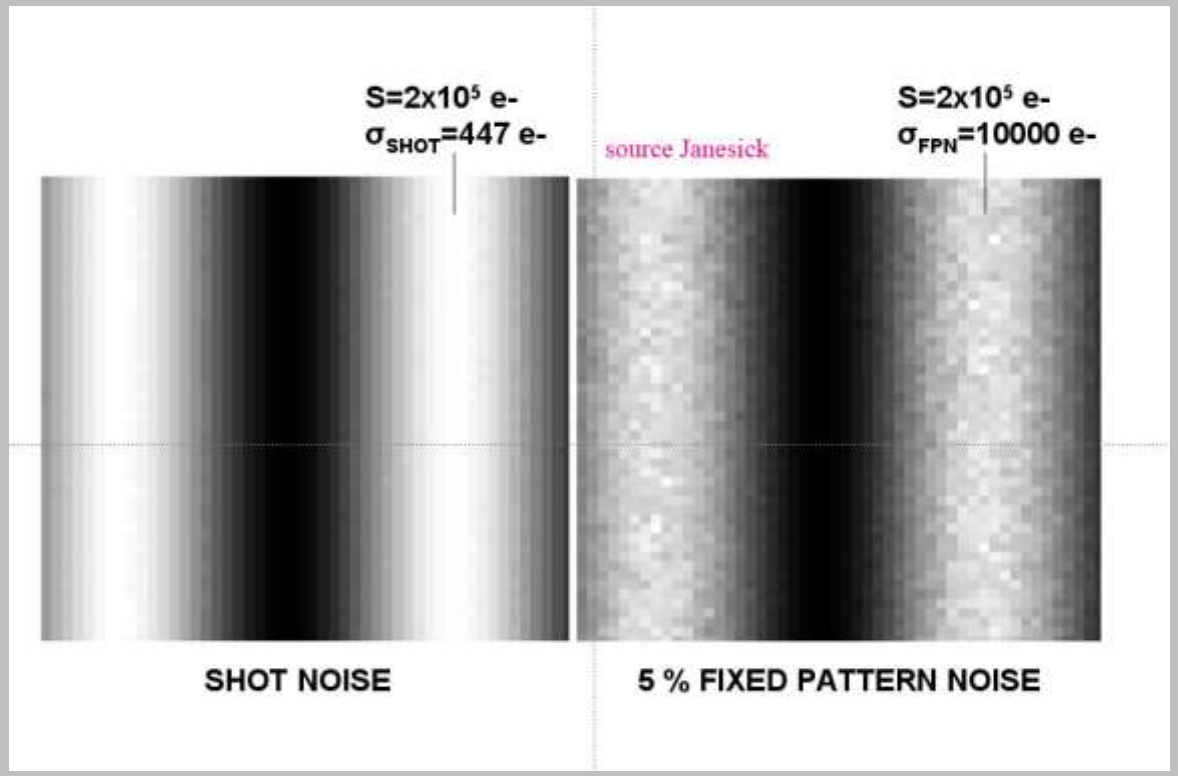
Speckle Noisy Image



image corrupted by Speckle Noise

4.Fixed Pattern





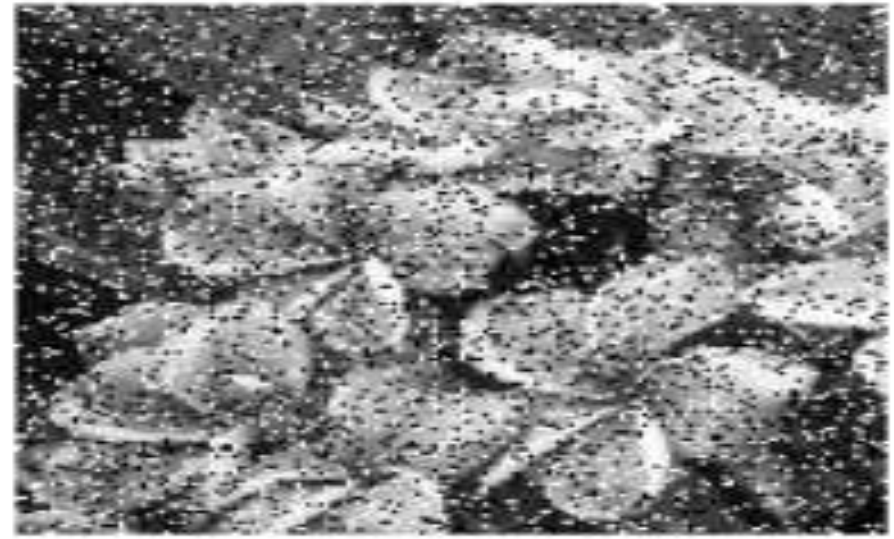


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Original image



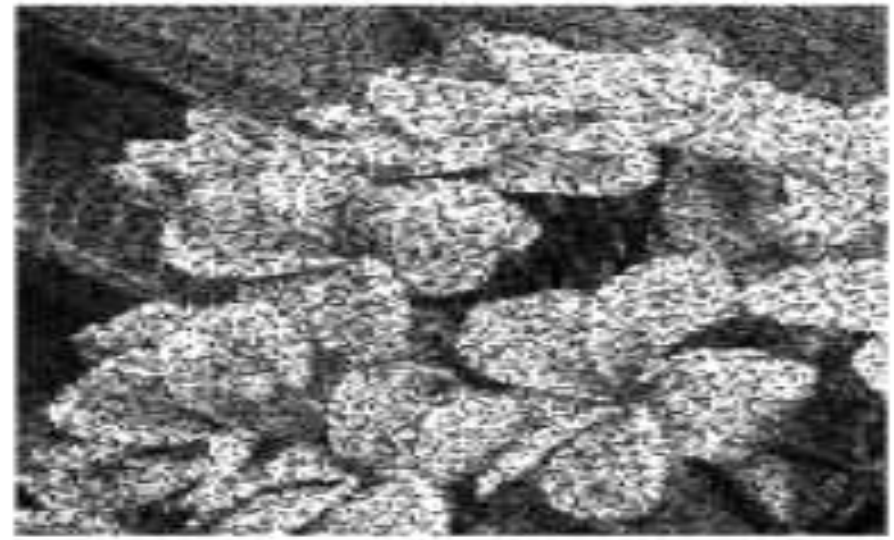
Salt and Pepper noise



Gaussian noise



Speckle noise



Denoising Kernels



Mean Filter

❑ Mean Filters:

- The value of each center pixel of the window is replaced by the average of all the values in the local neighbor within that particular window.
- The simplest Linear Filter. Suppose we apply 3x3 mean filtering

10	41	9
232	186	79
1	43	56



	73	

- Two 3x3 smoothing (averaging) filter mask

$\frac{1}{9}$ x

1	1	1
1	1	1
1	1	1

3x3 Mask for Average Filter

$\frac{1}{16}$ x

1	2	1
2	4	2
1	2	1

3x3 Mask for Weighted Average Filter

In a spatial averaging filter in which all the coefficients therefore sometimes it is known as a Box filter

Mean Filter (Contd..)

❑ Arithmetic Mean Filter:

- This filter removes local variations within the image.
- It is similar to the low-pass filter.
- It is useful in removing Gaussian Noise and Uniform Noise.
- It is a simple smoothing filter but it blurs the image.

$$\hat{f}(x,y) = \frac{1}{mn} \sum_{(x,y)\in S_{xy}} g(x,y)$$

Where,
g(x,y): Degraded image
S_{xy}: Set of coordinates in a rectangular window of size mxn

35	**45**	**57**	**128**	233	230
55	**178**	**255**	**255**	255	186
95	**65**	**78**	**190**	175	14
65	45	45	120	145	190
78	79	96	156	96	126
79	86	178	159	189	55

Degraded Image

*

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

Mask



35	**45**	**57**	**128**	233	230
55	**178**	**139**	**255**	255	186
95	**65**	**78**	**190**	175	14
65	45	45	120	145	190
78	79	96	156	96	126
79	86	178	159	189	55

Restored Image

Median Filter

- **Median Filter:**
  - It simply sorts the list and finds the median
  - The center pixel is replaced by the median value
  - It is an example of Non-linear filters
  - Excellent for removing Salt-and-Pepper Noise

$$\hat{f}(x,y) = \underset{x,y \in S_{xy}}{\text{median}}\{g(x,y)\}$$

35	**45**	**54**	**128**	233	230
55	**178**	**255**	**255**	255	186
95	**65**	**78**	**190**	175	14
65	45	45	120	145	190
78	79	96	156	96	126
79	86	178	159	189	55

Degraded Image

Median

(45,54,65,78,**128**,178,190,255,255)

35	**45**	**54**	**128**	233	230
55	**178**	**128**	**255**	255	186
95	**65**	**78**	**190**	175	14
65	45	45	120	145	190
78	79	96	156	96	126
79	86	178	159	189	55

Restored Image

Maximum and Minimum Filter

- Max filter is good for pepper noise and min is good for salt noise.

- **Max Filter:**

$$\hat{f}(x,y) = \max_{(s,t)\in S_{xy}} \{g(s,t)\}$$

- **Min Filter:**

$$\hat{f}(x,y) = \min_{(s,t)\in S_{xy}} \{g(s,t)\}$$

Gaussian Filter

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

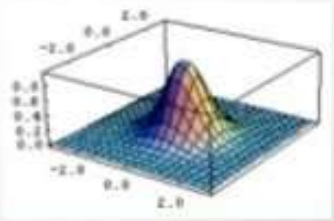
$F[x, y]$

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$H[u, v]$

This kernel is an approximation of a 2d Gaussian function:

$$h(u, v) = \frac{1}{2\pi\sigma^2} e^{-\frac{u^2+v^2}{\sigma^2}}$$



Gaussian kernels

1/16

1	2	1
2	4	2
1	2	1

1/273

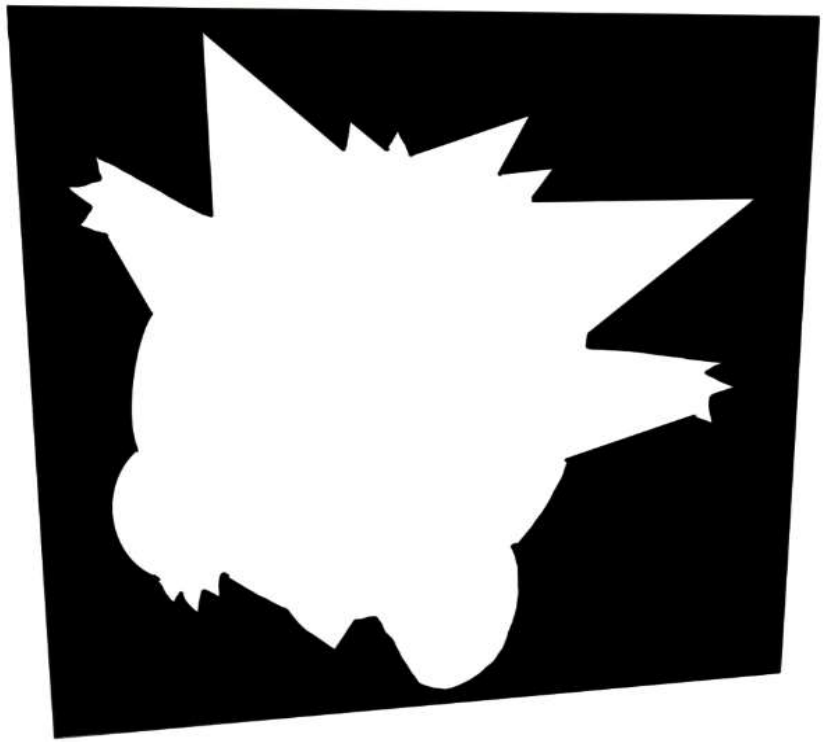
1	4	7	4	1
4	16	26	16	4
7	26	41	26	7
4	16	26	16	4
1	4	7	4	1

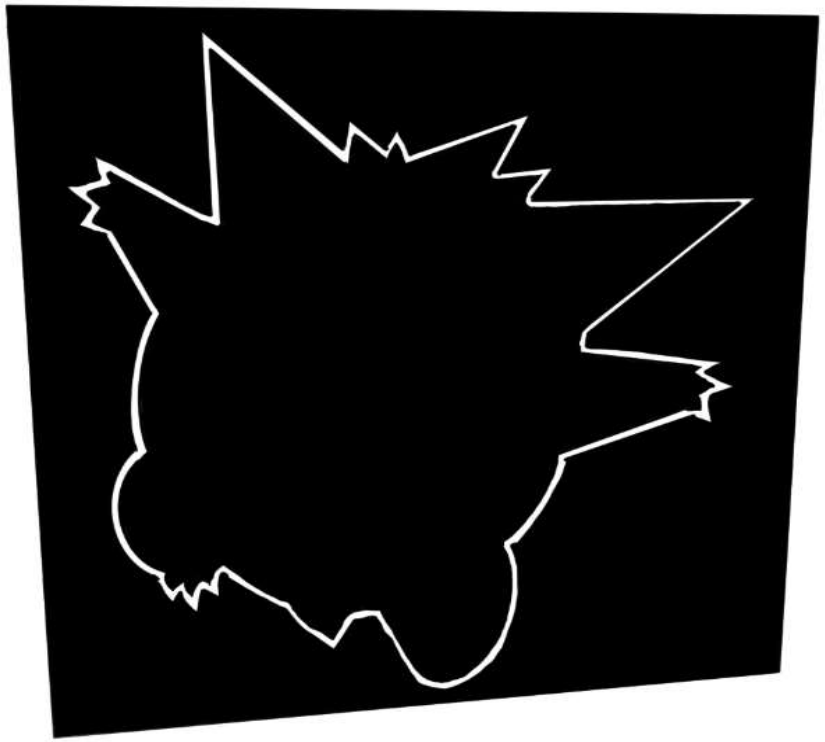
1/1003

0	0	1	2	1	0	0
0	3	13	22	13	3	0
1	13	59	97	59	13	1
2	22	97	159	97	22	2
1	13	59	97	59	13	1
0	3	13	22	13	3	0
0	0	1	2	1	0	0



Morphological Image Processing





Dilation / *Erosion* ?

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Dilation and Erosion



Erosion v/s Dilation

Erosion

- removal of structures of certain shape and size, given by SE
- Erosion can split apart joined objects and strip away extrusions

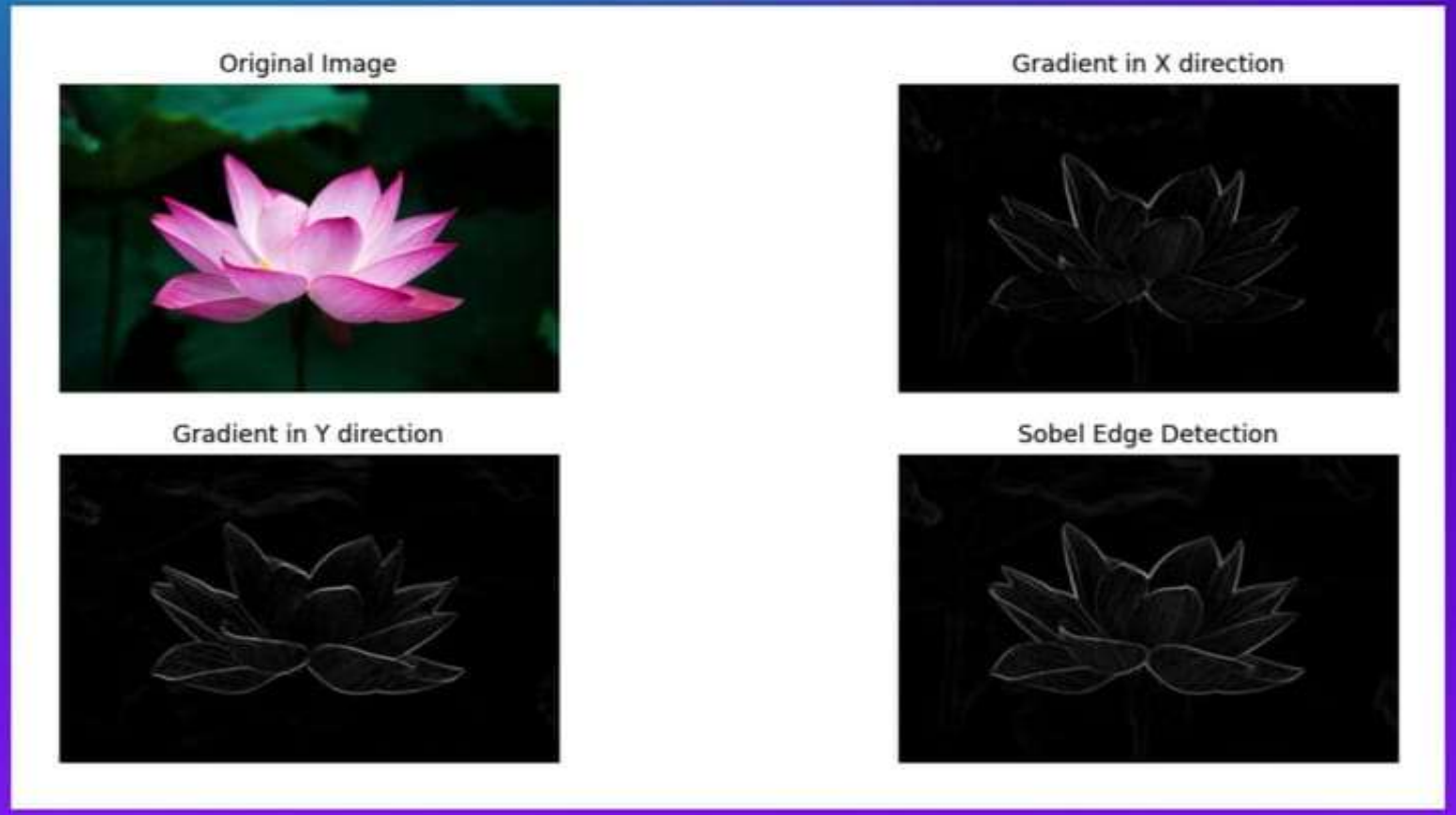
Dilation

- filling of holes of certain shape and size, given by SE
- can repair breaks and intrusions





Edge Detection in Image Processing



Original Image



Edge Image



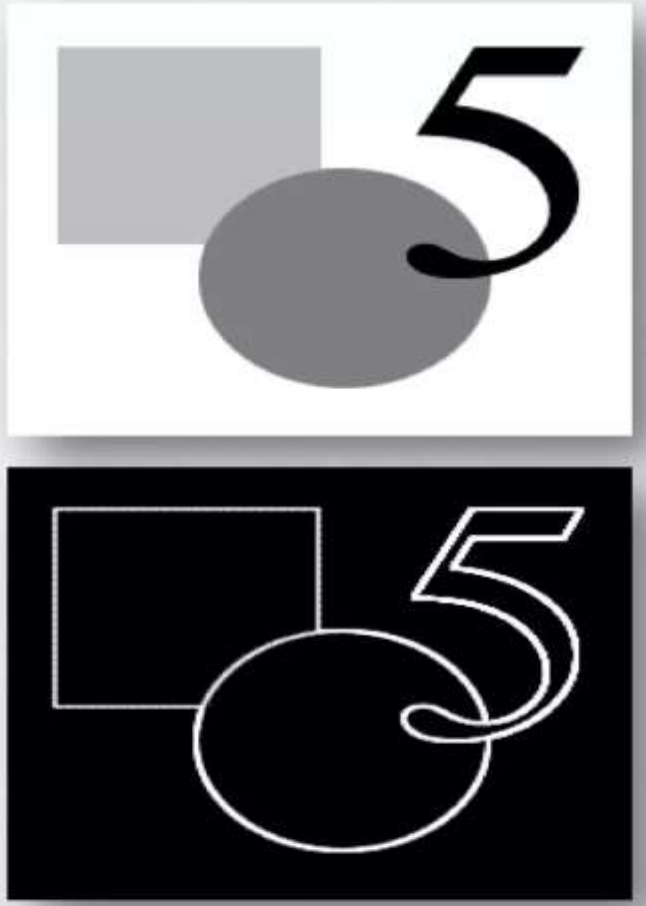
Edges

- Abrupt change in the intensity of pixels.
- Discontinuity in image brightness or contrast.
- Usually edges occur on the boundary of two regions .

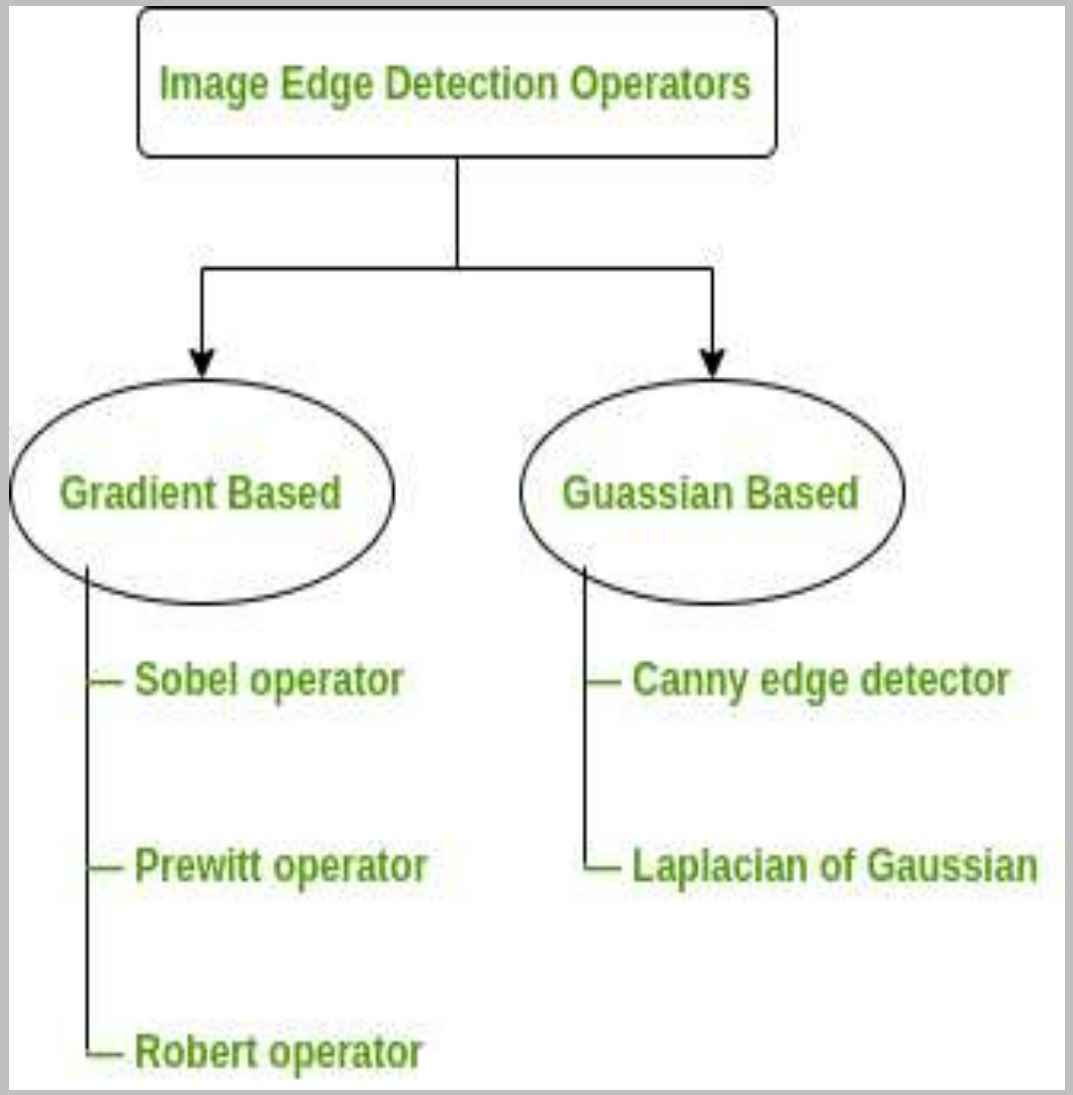


Edge Detection

- Process of identifying edges in an image to be used as a fundamental asset in image analysis.
- Locating areas with strong intensity contrasts.



Edge Detectors



1	0	0	1
0	-1	-1	0

Robert Mask

-1	0	1	1	1	1
-1	0	1	0	0	0
-1	0	1	-1	-1	-1

Prewitt Mask

-1	0	1	1	2	1
-2	0	2	0	0	0
-1	0	1	-1	-2	-1

Sobel Mask

Roberts

+1	0
0	-1

Gx

0	+1
-1	0

Gy

Original Image

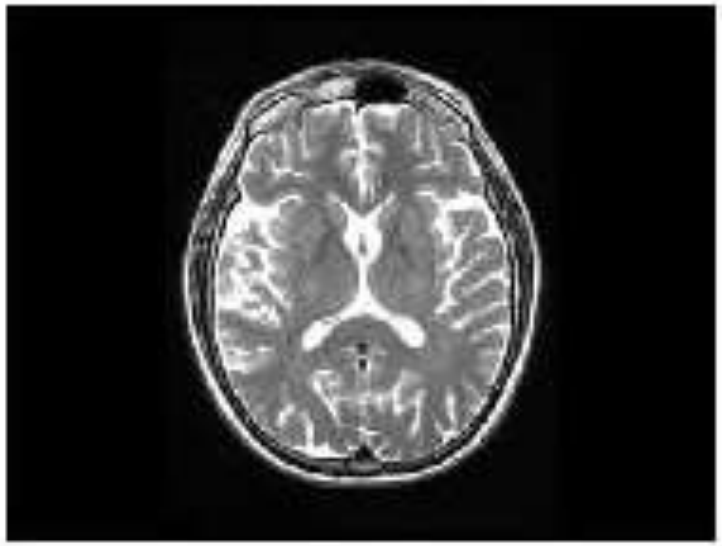


Image by Roberts Detector



Prewitt Edge Detection

Input Image:

10	9	9	4	0
0	6	6	2	2
5	9	8	4	3
7	5	5	4	3
8	10	8	5	0

Prewitt Kernels

-1	0	1
-1	0	1
-1	0	1

X-Direction

1	1	1
0	0	0
-1	-1	-1

Y-Direction

Sobel Edge Detection

Input Image:

10	9	9	4	0
0	6	6	2	2
5	9	8	4	3
7	5	5	4	3
8	10	8	5	0

Sobel Kernels

-1	0	1
-2	0	2
-1	0	1

X-Direction

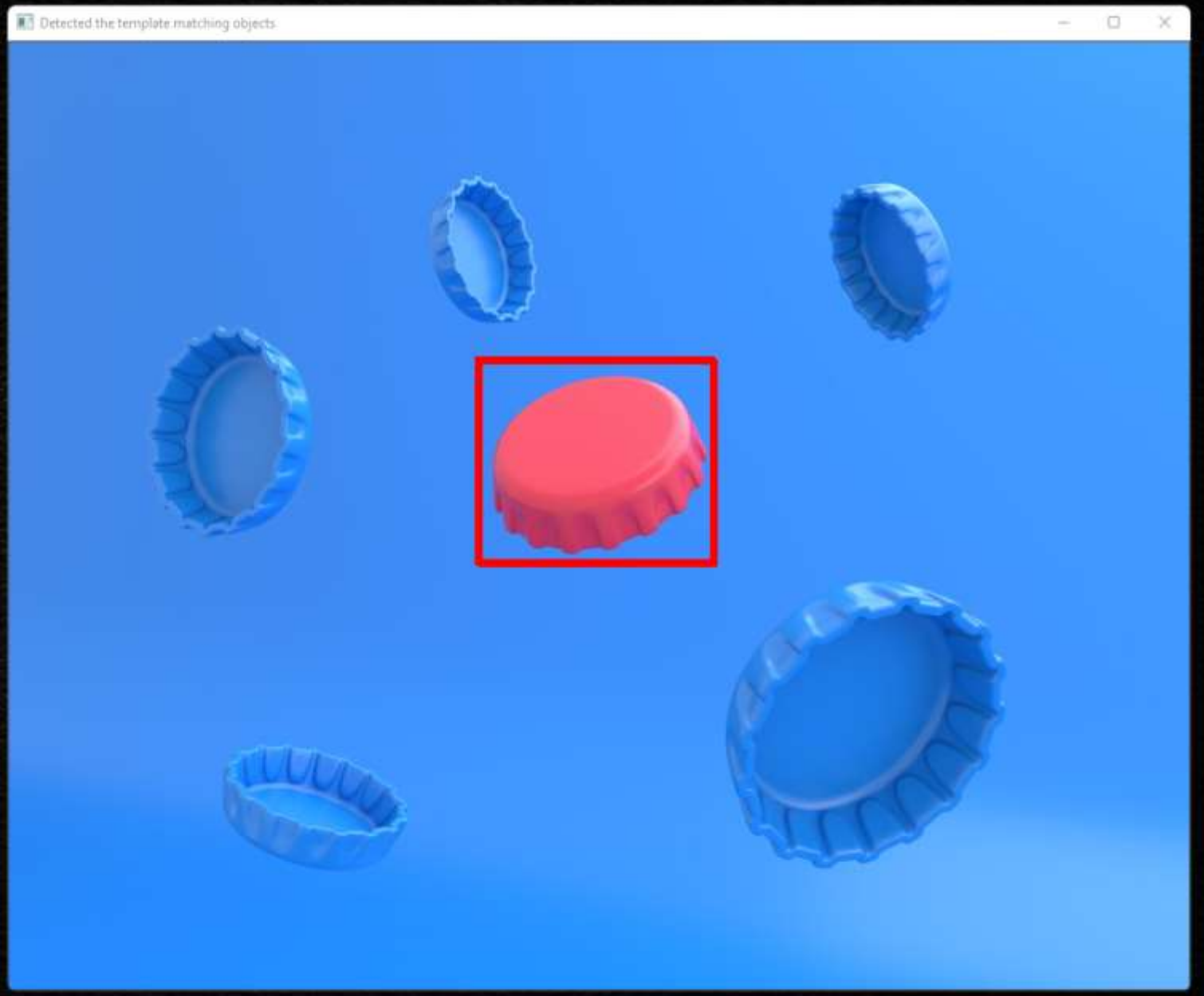
1	2	1
0	0	0
-1	-2	-1

Y-Direction

Template Matching



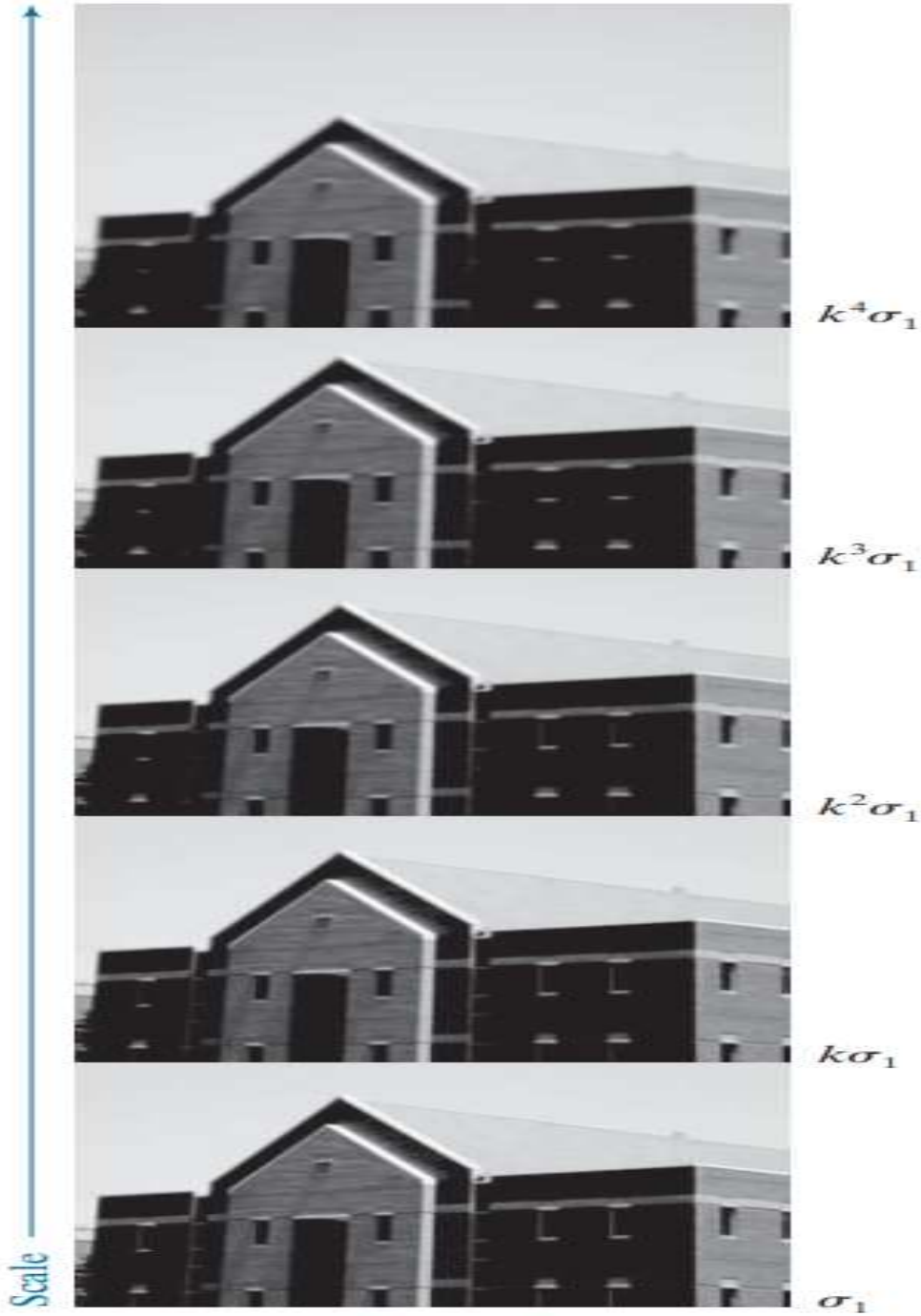


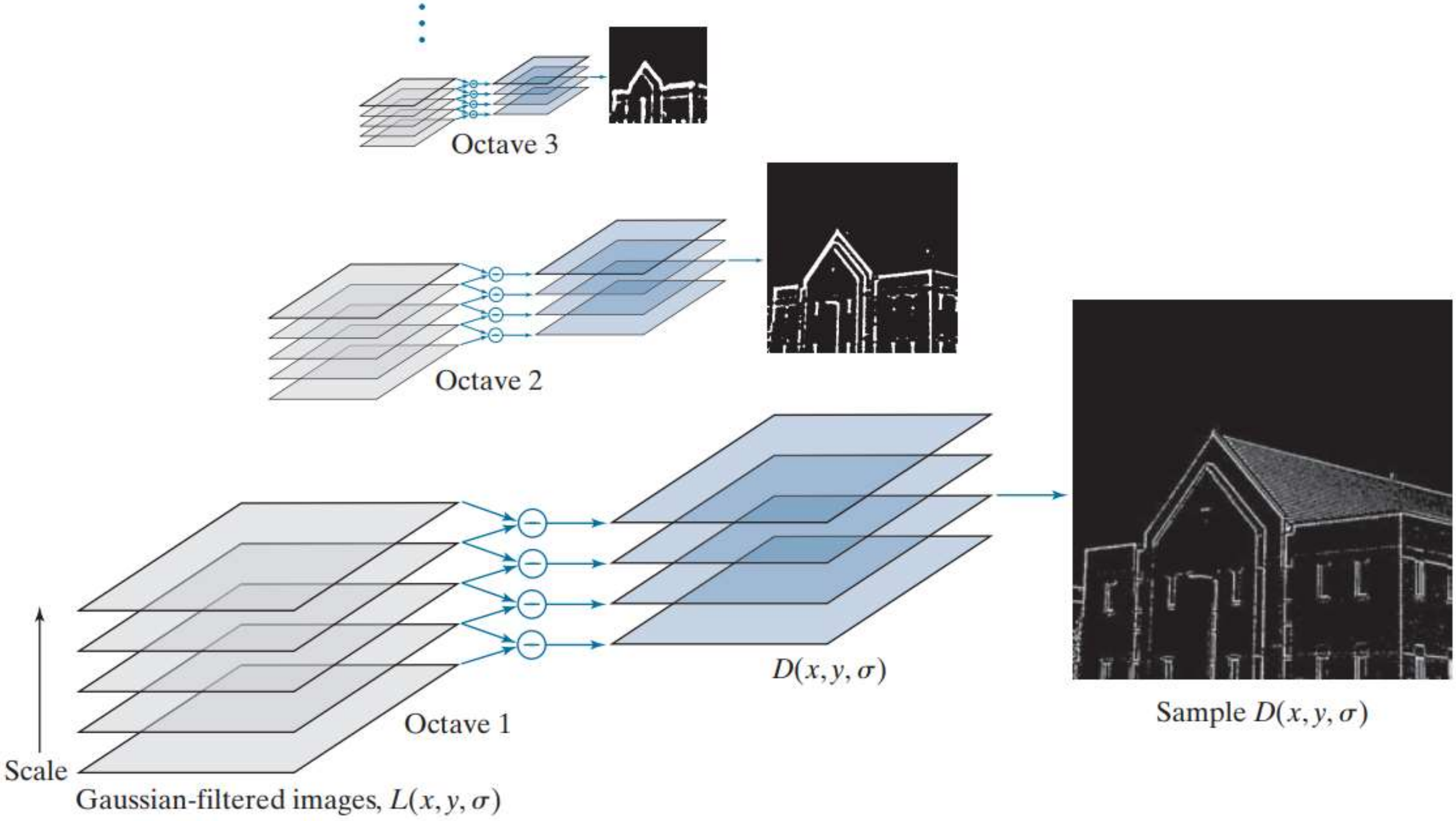


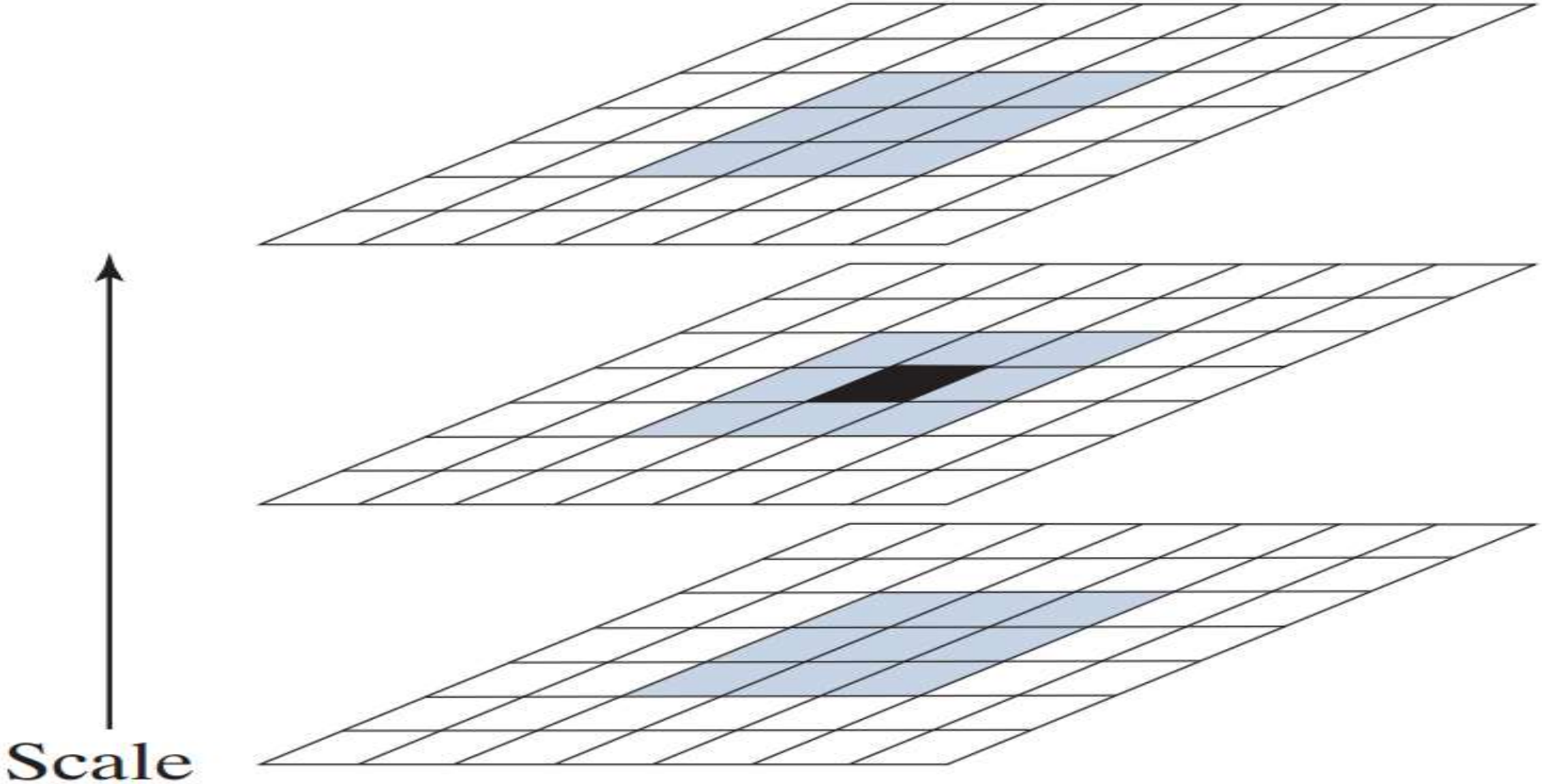
What is Scale-Invariant Feature Transform (SIFT)?

roboflow




Scale
$k^4\sigma_1$
$k^3\sigma_1$
$k^2\sigma_1$
$k\sigma_1$
σ_1
Octave 1
Scale
$k^4\sigma_2$
$k^3\sigma_2$
$k^2\sigma_2$
$k\sigma_2$
$\sigma_2 = 2\sigma_1$
Octave 2
Scale
$k^4\sigma_3$
$k^3\sigma_3$
$k^2\sigma_3$
$k\sigma_3$
$\sigma_3 = 2\sigma_2 = 4\sigma_1$
Octave 3


Octave 3
Octave 2
Octave 1
$D(x, y, \sigma)$
Sample $D(x, y, \sigma)$
Scale
Gaussian-filtered images, $L(x, y, \sigma)$



Corresponding sections of three contiguous $D(x, y, \sigma)$ images



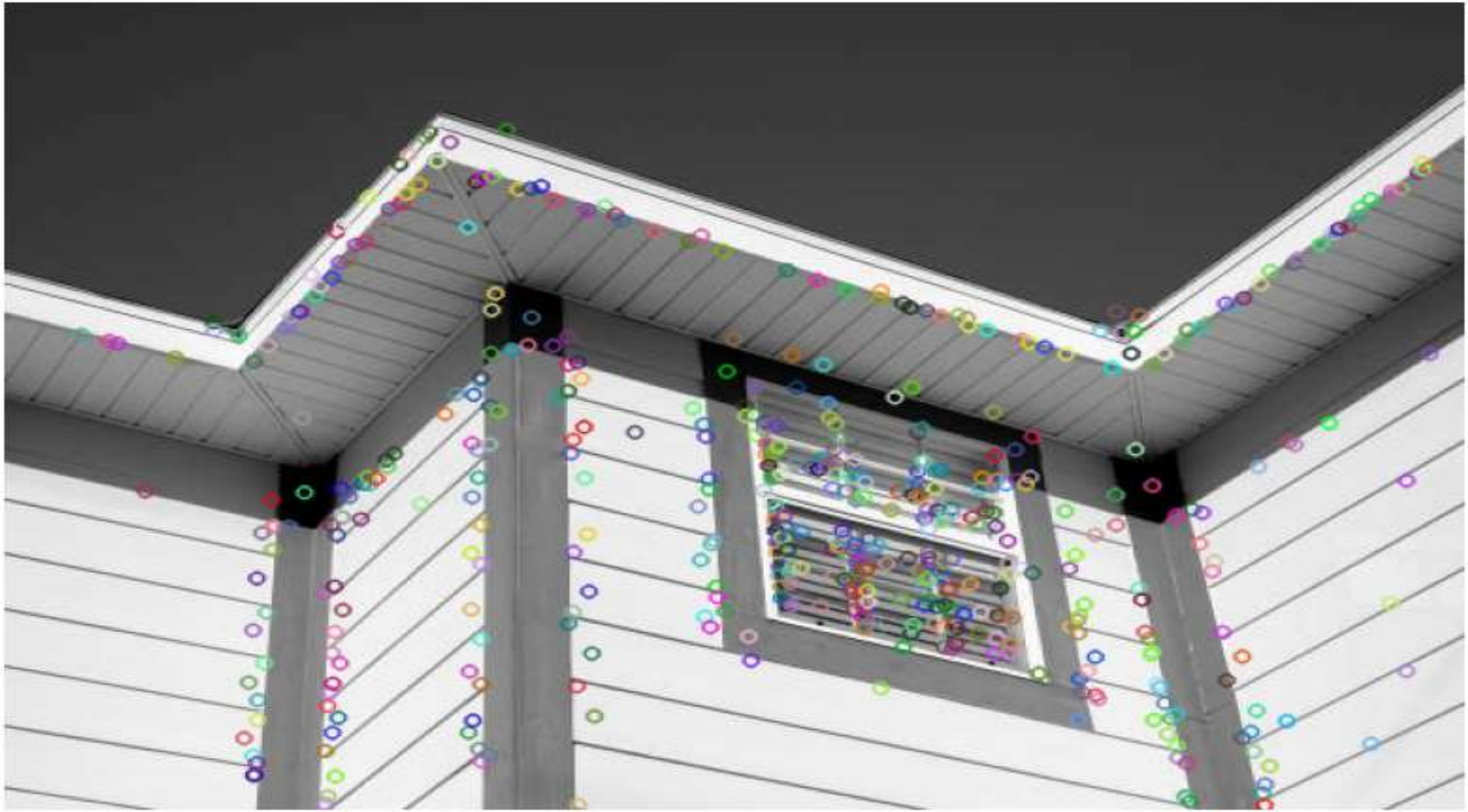
(a)



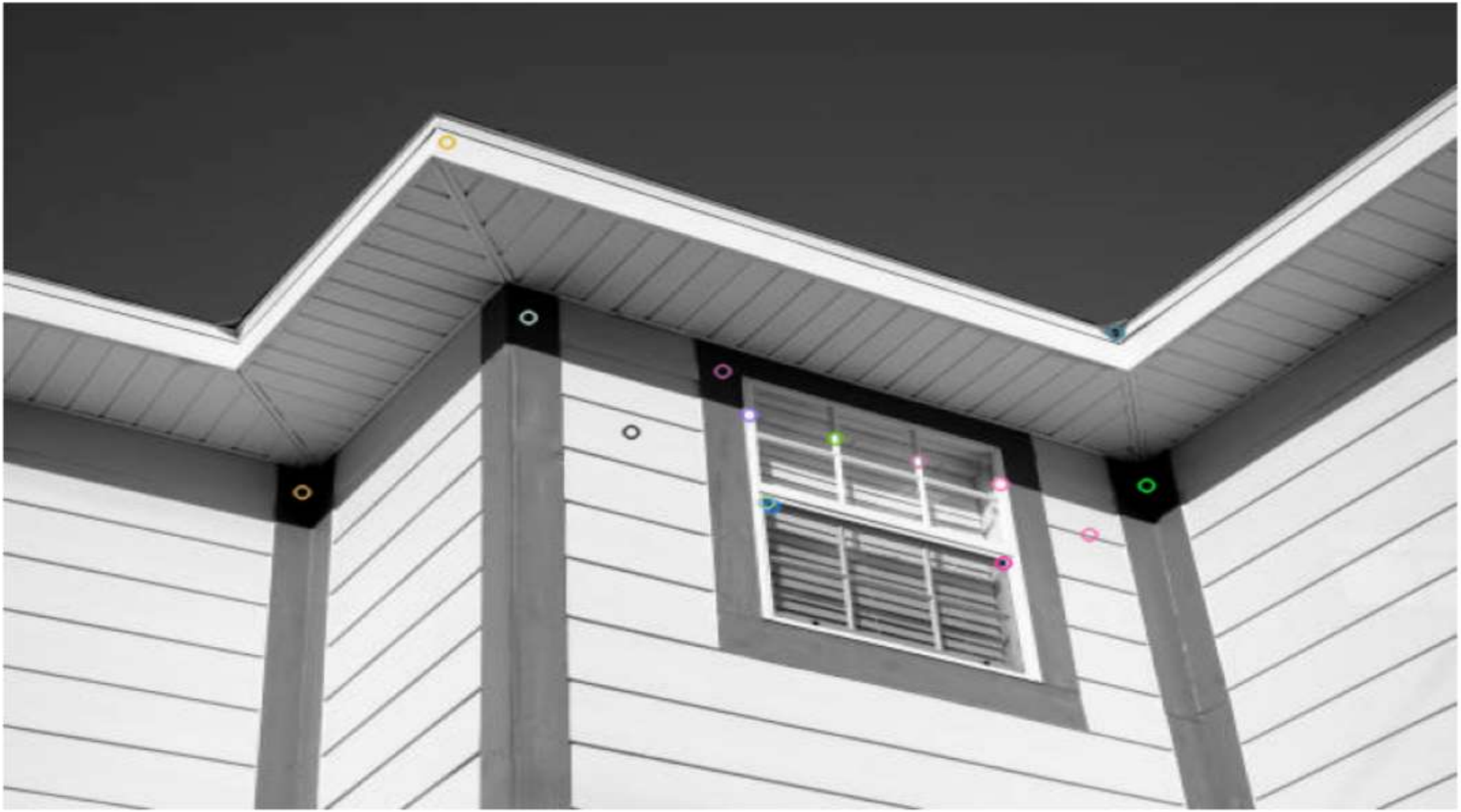
(b)



(c)





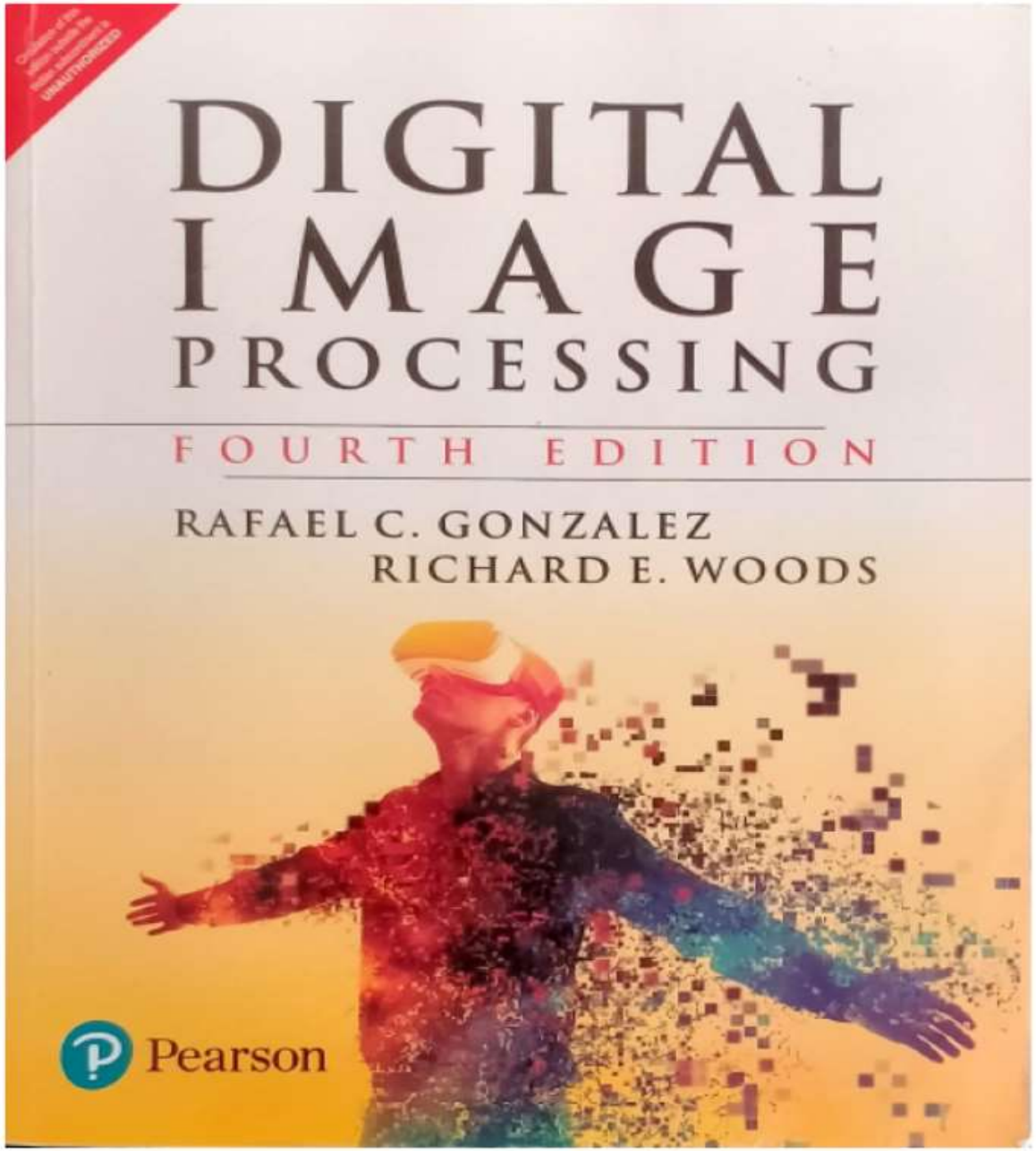


```
{
  "status": "PASS",
  "keypoints_detected": 172
}
```

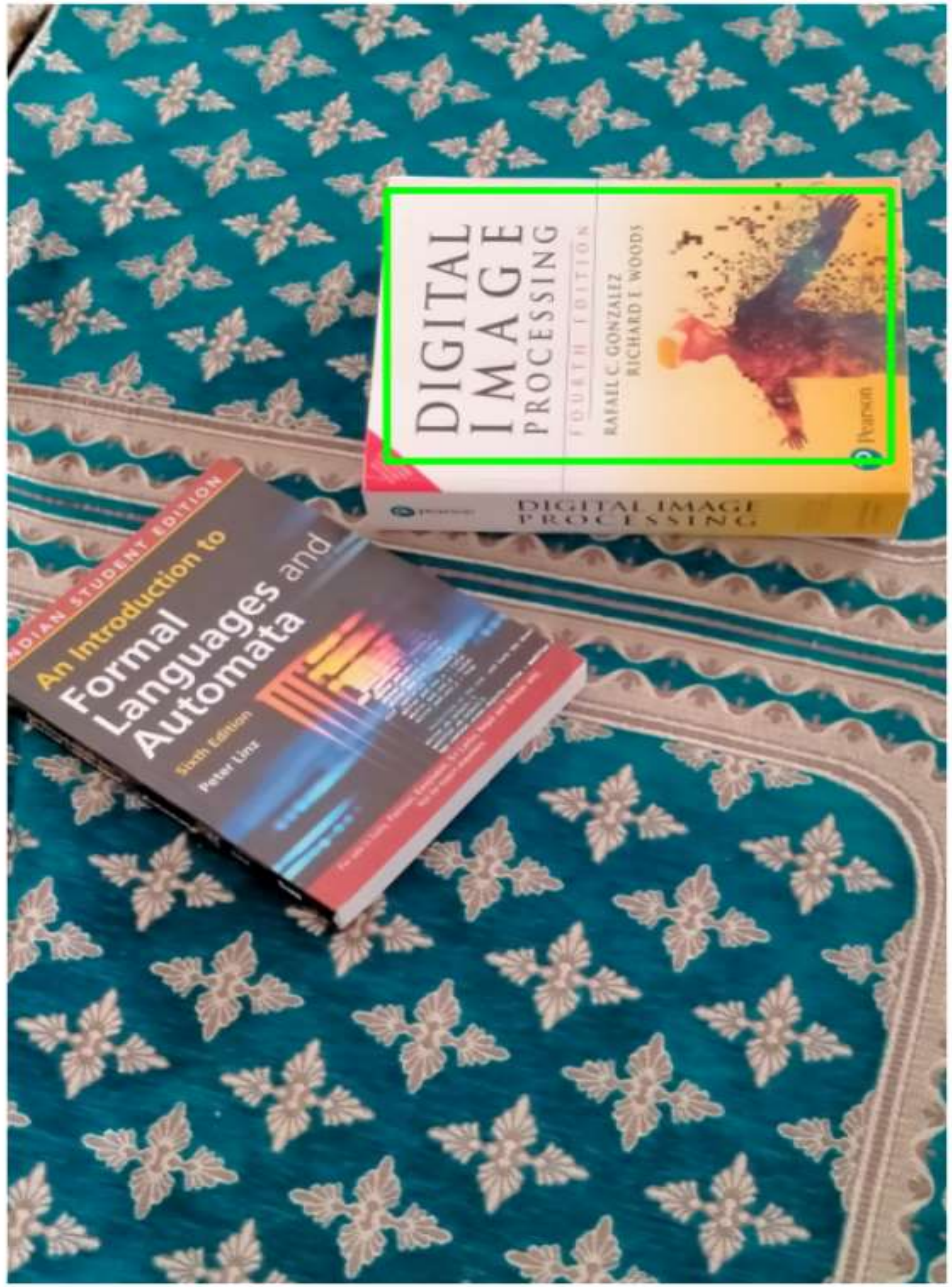

Result: PASS (172 keypoints)



Input Object



Detection Result: Object Found
(248 matches)




```
Status: Object Found
Number of good matches: 248
```

Thank You

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